

The Hodge Traversal, Compiled Edition

A registry-charted semantic traversal of the Hodge conjecture problem space, with instrument completion theorems, literature welds, and exhaustion at the corrected stopping condition

Versions V4–V8, unified · Lu Semita · EmergenceByDesign — co-authors to be named upon final formalization

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Preface: an invitation to refutation

This document is distributed with a standing, explicit invitation: **find the hole**. It is written to be defeated if it is defeatable, and every claim in it is tagged with the exact standard against which it should be attacked. The authors regard a successful refutation of any component as a contribution to the work, to be incorporated by name as a node update in the next version.

To defeat this document, a reader must exhibit at least one of the following. Each is a precise target; vague dissatisfaction with framing does not score.

R1 — A false Tier-I statement. Every sentence labeled Tier I is asserted as standard, checkable mathematics or as an accurate citation of the published literature. Exhibit one that is false as stated.

R2 — An invalid inference in a Tier-II proof. The Tier-II results (the eigenspace obstruction, the non-separating certificate theorem, the residue trichotomy, the completion theorem, the verdict-lattice and forcing-wall applications) are proved. Exhibit a step that does not follow.

R3 — A claim exceeding its declared tier. The document's own discipline forbids any sentence from asserting more than its tier permits. Exhibit a sentence whose content, read plainly, outruns its label — in particular, any sentence that asserts the truth, falsity, or formal independence of the Hodge conjecture.

R4 — A residue-decreasing operation inside the declared instrument. Theorem 20.1 asserts that no operation in the declared language $\mathfrak{L}$ — conjugation-fixed reattachments, Hodge-necessary certificates, cycle-unrealized structure morphisms, closed under composition and the stated reframings — decreases any residue. Exhibit such an operation and the decrease.

R5 — An existing violable filter. Section 26 asserts that every necessary condition of algebraicity available in current mathematics (rationality, Hodge type, harmonicity, absolute-Hodgeness, motivated-ness) is satisfied by the

surviving target classes, so that no refutation-by-example of the Hodge conjecture is derivable from any existing filter. Exhibit a currently existing necessary condition that a surviving target violates.

R6 — A breach of the forcing wall. Section 25 asserts, conditional on the standard spreading-out formalization placing the Hodge statement at Shoenfield- absolute complexity, that forcing cannot establish its independence from ZFC. Exhibit either a formalization that legitimately escapes the pointclass, or an error in the absoluteness application.

R7 — A counterexample to the corrected verdict lattice. Section 24 asserts, with $\text{Con}(\text{ZFC})$ as witness, that "unprovable" does not entail "provably unprovable" and that no consistent recursively axiomatized system certifies all of its own gaps. Exhibit an error in that classical argument or in its application here.

Absent a successful exhibit under R1–R7, the document's scoped results stand as proved, in the ordinary sense in which mathematics is proved: the burden is now on the reader, and the reading instructions are complete.

What this document actually proves, in the standard academic sense

Stated once, maximally, and exactly — this is the strongest claim the mathematics supports, and the document claims all of it and none beyond it.

1. Unconditional limitation theorems for a declared method family. The document defines an explicit class of approaches to the Hodge conjecture — every conjugation-invariant algebraic reattachment, every certificate whose acceptance is implied by Hodge membership, every Hodge-structure morphism unaccompanied by a cycle, closed under composition and under change of CM field, discriminant, dimension, operator, and carrier-internal reframing — and **proves, at full and elementary rigor, that no member of this class, iterated finitely or infinitely, can decide the algebraicity of a conjugation-anti-invariant Weil-type Hodge class, in either direction** (Theorems 17.4, 17.6, 18.2, 20.1). These results are of the same established genre as restricted proof-system lower bounds: unconditional theorems about what a specified family of methods cannot do, silent about stronger methods, and fully falsifiable under R2 and R4.

2. A correct and current map of the frontier. The document accurately records (Tier I, cited) that the Weil fourfold regime — the traversal's original open core — has been closed affirmatively by Markman and successors for all imaginary quadratic fields and all discriminants; that the Kuga–Satake correspondence is now algebraic on the generalized-Kummer locus and on infinitely many K3 families; that the open CM frontier has relocated to abelian sixfolds of Weil type outside discriminant -1 , with a named rigid family, proven absolute Hodge and proven inaccessible to all currently known methods, standing as the sharpest residue and carrying a pending finite computation. The closure mechanism — carrier change into hyper-Kähler geometry with deformation-stable algebraic sheaf classes — is observed to instantiate exactly the weld specification the instrument had priced, which is the strongest form of validation a limitation analysis can receive: its wall was crossed precisely where, and precisely how, it said crossing would require.

3. Correct classical-logic bounds on the demand for decidability. The document proves, by correct application of Gödel's second incompleteness theorem, that the demand "every unprovable statement is provably unprovable" is false in general — refuted with the exhibited witness $\text{Con}(\text{ZFC})$ — so that a fourth verdict cell, *open*, is ineliminable from any honest stopping condition; and it establishes, conditional only on the standard complexity reading of the spread-out Hodge statement, that Shoenfield absoluteness bars the forcing route to any independence result for the Hodge conjecture, while recording (Tier II, empirical over the corpus) that no consistency-strength attachment exists or has been proposed.

4. The exact specification of every remaining exit. The document reduces all possible future resolutions of the examined frontier to a finite list of door-specifications (Remark 20.3, Sections 26–27), one of which has since been opened by the literature in the stated form, and states — as its single strongest synthetic finding — that **at the present state of mathematics the Hodge conjecture is not proven, is not refutable by any existing necessary condition, and is not addressable by the only general independence machinery set theory possesses**; it therefore occupies, demonstrably and not by default, the open cell of the corrected verdict lattice, with its live frontier consisting of named objects and stated finite tasks, and with every recorded movement of the modern era running in the algebraicity-ward direction.

What is not claimed. The document does not prove, and does not claim to prove, the Hodge conjecture, its negation, or its formal independence of any system. No sentence in it may be quoted to that effect without misquotation, and Section 28's compression rules forbid the three specific inflations by which such a misreading would be manufactured. Whether the scoped results above merit standing as a contribution is for the mathematical community to judge; the document's posture is that the judgment should proceed through R1–R7, and it will be answerable to the outcome.

PART ONE — The Traversal (V4)

Sections 1–16: protocol, objects, boundary nodes, the Weil/CM regime, K3 and Kuga–Satake, scope widening, the three-shape frontier, conclusive scoped status, and the outside-frame transcendence traversal. Node statuses in this Part reflect the state of the traversal as originally executed; Part Three records the literature events that have since closed several nodes. The original text is preserved so that the instrument's predictions can be checked against what subsequently occurred.

Abstract

This document presents a semantic traversal of selected Hodge-theoretic problem spaces. It does not claim to prove, refute, or decide the Hodge conjecture. Its purpose is more limited and more precise: to define an explicit family of objects, bridges, predicates, and stopping conditions; to push each bridge until it either closes or produces a definable residue; and to record the resulting frontier without inflating bridge failure into a universal statement about all possible mathematics.

The traversal distinguishes co-location from weld. A registry may locate two structures at a common address or in an adjacent conceptual register, but such co-location does not transfer truth. Transfer requires a weld: a theorem or construction that rigidly couples the frames so that information in one frame forces information in the other. This distinction controls the entire analysis.

The result is a three-shape boundary for the portion of the Hodge problem space examined here. Divisor-level cases, such as degree-two Hodge classes on K3 surfaces, close by Lefschetz (1,1). CM anti-invariant Weil-type classes expose a no-weld regime: norm-type reattachments land in the conjugation-invariant component, while spectral or harmonic data certify the wrong predicate. Kuga–Satake constructions provide a third regime in which a genuine Hodge-structure weld exists, but the algebraicity of the corresponding cycle remains open in general. A final outside-frame analysis examines absolute Hodge theory and period-transcendence methods. These outside frames do not establish Hodge-non-algebraicity; the genuine Lindemann-style transcendence machinery concerns periods, not rational Hodge classes as cycle classes.

The traversal therefore earns only a scoped conclusion: within the defined bridge family and the tested outside frames, the transcendence reading does not cross the Hodge predicate, while the algebraicity question remains open.

Posture and scope

This is a semantic traversal, not a proof attempt. The goal is to define the problem space of the Hodge conjecture in decidable language, push available bridges until each either closes or generates a definable residue, and log the result so that the conclusion emerges from the completed description rather than from assertion.

No claim is made or implied about the truth, falsity, or provability of the Hodge conjecture. Where a bridge stops, the stop is a statement about that bridge's reach within its stated frame, not about the existence or non-existence of algebraic cycles. The Hopf registry appears here only as a declared chart: a coordinate system for locating and comparing frames. It does not function as a generative mechanism, a truth oracle, or a substitute for a theorem.

The following tier labels govern the document.

Tier I — Verifiable. Standard, checkable mathematics, stated as fact.

Tier II — Structural. Exact correspondences and classifications read within the framework after the Tier-I objects have been fixed.

Tier III — Programmatic. Forward-looking claims, conjectural directions, and author-defined interpretive structures. These are not asserted as established mathematics.

1. Traversal protocol

The traversal follows a fixed protocol so that termination is a design property rather than an aspiration.

1. **Initial condition.** Each node begins from an explicitly stated mathematical object and target predicate.
2. **Finite reduction.** From the stated initial condition, admissible reductions proceed finitely unless the reduction generates an irreducible residue.
3. **Residue as definable stop.** An irreducible residue is a point at which a reduction cannot continue inside the stated frame. When that point is detectable and nameable, it becomes a recorded stopping condition rather than an unbounded recursion.
4. **Two stop types.** An *exhaustion stop* occurs when every condition at the node reduces to decidable language; the branch closes and its verdict is recorded. An *irreducible-residue stop* occurs when the stated frame cannot carry the reduction further; the residue's coordinate and type are recorded, the node is marked open relative to that frame, and the traversal departs.
5. **Depart and do not re-enter.** A logged node is not revisited unless a new bridge is explicitly added to the scope. Ground already exhausted is not re-covered.

This mirrors the Dynamical Semantic Traversal cycle: reduce until marginal structural return is non-positive, record the residue with its coordinate, and depart.

The word *irreducible* always means irreducible relative to the stated bridge, frame, and predicate. It does not mean intrinsically irreducible relative to all possible mathematics.

2. Objects and definitions

2.1 The Hodge conjecture

Let X be a smooth projective variety over $\mathbb{C}$. For each k , the rational Hodge classes are

$$\text{Hdg}^k(X) = H^{\wedge\{2k\}}(X, \mathbb{Q}) \cap H^{\wedge\{k,k\}}(X).$$

An algebraic cycle of codimension k is a finite $\mathbb{Q}$ -combination of codimension- k subvarieties. Its cohomology class lies in $\text{Hdg}^k(X)$. The Hodge conjecture asserts the reverse inclusion: every rational Hodge class is the class of an algebraic cycle.

2.2 Residue lens

For $\alpha \in \text{Hdg}^k(X)$ define the residue notation

$\mathcal{J}(\alpha)$ = obstruction of α to lying in the $\mathbb{Q}$ -span of algebraic-cycle classes.

This notation does not solve the Hodge question. It records where the open predicate sits. One has

$$\mathcal{J}(\alpha) = 0 \iff \alpha \text{ is algebraic.}$$

Thus deciding $\mathcal{J}(\alpha) = 0$ is logically identical to deciding the Hodge question for α . The residue lens is bookkeeping, not a reduction that removes the difficulty.

2.3 Registry as chart

A Hopf configuration

$$S^1 \hookrightarrow S^{\wedge\{2n+1\}} \rightarrow \mathbb{CP}^n$$

is used as a declared scaffold for an extendable family of circumscribing shells. Its role here is strictly chart-like: it assigns frames such as algebraic, spectral, and arithmetic frames stable addresses and makes adjacency between frames statable.

By the registry/map separation, a descriptively complete registry is inert as a truth oracle. A chart locates and relates; it does not decide. Two notions remain distinct throughout:

- **Co-location:** two frames sit at the same chart address or in a statable adjacency relation.
- **Weld:** a theorem, construction, or correspondence rigidly coupling two frames so that a fact in one frame forces a fact in the other.

Co-location may be necessary for a weld, but it is not sufficient.

3. Boundary nodes

3.1 Projective space: $\mathbb{CP}^n$

Tier I. The cohomology ring of complex projective space is

$$H^*(\mathbb{CP}^n) = \mathbb{Z}[\omega]/(\omega^{n+1}),$$

with ω the hyperplane class. Every cohomology class is an integer multiple of a power ω^k , and ω^k is the class of a linear subspace $\mathbb{CP}^{n-k}$. Hence the Hodge conjecture holds on $\mathbb{CP}^n$ trivially.

Stop. Exhaustion. Matching spectral or registry data to powers of ω engages none of the conjecture's difficult content. Projective space is a closed boundary node, not the frontier.

3.2 The Hopf total space as category constraint

Tier I. The sphere S^{2n+1} has odd real dimension and rational cohomology only in degrees 0 and $2n+1$. It is not a complex projective variety and does not carry a Hodge structure of the type required for the Hodge conjecture. The map $S^{2n+1} \rightarrow \mathbb{CP}^n$ is a smooth $U(1)$ -bundle, not a morphism of projective varieties.

Consequence. A literal instruction to lift an algebraic class to the Hopf total space has no referent in the category where algebraic cycles live. This does not prevent the chart use of the Hopf configuration. Co-location remains stable. It prevents only the stronger reading in which the shell itself functions as an algebraic carrier.

4. Node 0: abelian threefold $E_1 \times E_2 \times E_3$

Initial condition. Let $A = E_1 \times E_2 \times E_3$ be a product of three elliptic curves. Then $\dim_{\mathbb{C}} A = 3$. The target is the degree-four Hodge classes.

Tier I. The Hodge numbers of a product of three elliptic curves satisfy $h^{p,q} = C(3,p) \cdot C(3,q)$. Thus $h^{2,2} = C(3,2)^2 = 9$, with the standard Hodge diamond for such a product.

For a generic product with no extra isogenies or complex multiplication, the rational Hodge classes are generated by divisor classes and are therefore algebraic. The genuinely open non-divisor Hodge classes do not appear on a generic threefold of this type.

Stop. Exhaustion. The node is closed and departed.

Frontier relocation. The canonical home of non-divisor Weil classes lies in abelian fourfolds with complex multiplication by an imaginary quadratic field, as in Weil's construction. The Hodge conjecture for these classes remained open in general at the time of this Part's writing, with scattered special cases resolved. (*Part Three records the subsequent closure of this regime.*)

Tier II chart note. An imaginary quadratic CM field $\mathbb{Q}(\sqrt{-d})$ occupies the negative-discriminant register of the same quadratic-field structure that appears in Gaussian-period factorizations. For primes $p \equiv 3 \pmod{4}$, the sign convention $p^* = (-1)^{(p-1)/2} p < 0$ gives the negative register. This is recorded as co-location, not as a weld.

5. Node 1: Weil fourfold with CM by $\mathbb{Q}(\sqrt{-3})$

5.1 Initial condition and target

Let A be an abelian fourfold of Weil type with complex multiplication by $K = \mathbb{Q}(\sqrt{-3})$. Then $\dim_{\mathbb{C}} A = 4$, and the Hodge numbers satisfy $h^{\{p,q\}} = C(4,p) \cdot C(4,q)$. The middle values include

$$h^{\{2,0\}} = 6, h^{\{2,1\}} = 24, h^{\{2,2\}} = C(4,2)^2 = 36.$$

Inside $H^{\{2,2\}}(A)$ sits a distinguished subspace $W_A \subset \text{Hdg}^2(A)$ of Weil classes. These classes are characterized as a Galois-anti-invariant eigenspace of the K -action on cohomology and are not contained in the subring generated by divisor classes. The target is precisely this non-divisor component.

5.2 Bridge A: norm-cycle reattachment

The programmatic algebraic reattachment considered here uses a norm $N(g) = g\bar{g}$, the product of an element with its Galois conjugate over K . The norm is, by construction, invariant under $\text{Gal}(K/\mathbb{Q})$. Hence it produces classes in the Galois-fixed component.

The target Weil classes occupy the Galois-anti-invariant component. The norm therefore lands in the wrong eigenspace. It cannot produce the target classes by this construction.

Stop. Irreducible residue relative to Bridge A.

- Type: frame mismatch.
- Coordinate: the Galois-anti-invariant eigenspace of the K -action on $H^{\{2,2\}}(A)$.
- Missing weld: a reattachment operation preserving the anti-invariant component and realized by an algebraic cycle.

This records only the reach of the norm bridge. It does not assert that no algebraic cycle exists in W_A .

5.3 Bridge B: Beltrami / Dirac spectral bridge

The abelian variety A is compact Kähler. By Hodge theory, every class in $H^{\{2,2\}}(A, \mathbb{Q})$ has a unique harmonic representative. Laplace–Beltrami and Dirac-type spectral data can certify harmonicity. However, harmonicity is automatic for Hodge classes. The open predicate is algebraicity, not harmonicity.

Thus spectral membership carries no information that separates algebraic Hodge classes from non-algebraic Hodge classes within the Hodge locus. The spectral bridge resolves the wrong predicate.

Stop. Irreducible residue relative to Bridge B.

- Type: predicate mismatch.
- Coordinate: the gap between harmonic representative and algebraic cycle.
- Missing weld: a theorem coupling a spectral invariant to the cycle-class predicate.

5.4 Node status

Both bridges stop at the same seam from opposite directions. Algebraicity is a Galois-rational cycle-class property. The norm bridge reaches the conjugation-invariant component. The spectral bridge certifies harmonicity, which all Hodge classes already possess. Neither bridge reaches the Galois-anti-invariant Weil component as an algebraic cycle.

Node status. Departed with two logged residues. (*Closed affirmatively by external theorem; see Part Three, Section 22.*)

6. Node 2: Weil fourfold with CM by $\mathbb{Q}(\sqrt{-7})$

This node tests whether the residue types from Node 1 depend on the specific CM field.

Let A be an abelian fourfold of Weil type with CM by $K = \mathbb{Q}(\sqrt{-7})$. The field has class number one and occupies the negative register associated with the $p = 7$ Gaussian-period atom.

For Bridge A, the norm $N(g) = g\bar{g}$ remains invariant under $\text{Gal}(K/\mathbb{Q})$ for every imaginary quadratic field. Weil classes remain conjugation-anti-invariant. Therefore the same frame-mismatch residue recurs, with only the field coordinate changed.

For Bridge B, compact Kähler geometry again supplies harmonic representatives for all Hodge classes. Spectral data certify harmonicity, not algebraicity. The same predicate-mismatch residue recurs.

Result. The prediction is confirmed. The residue types do not depend on the discriminant d ; the field changes the coordinate, not the residue mechanism. (*Closed affirmatively by external theorem; see Part Three, Section 22.*)

7. Node 3: K3 surface in degree two

7.1 Initial condition

Let X be a K3 surface. The target is the degree-two Hodge classes.

Tier I. The second integral cohomology lattice is

$$H^2(X, \mathbb{Z}) \cong U^3 \oplus E_8(-1)^2,$$

with signature $(3, 19)$. The Hodge numbers are $(h^{\{2,0\}}, h^{\{1,1\}}, h^{\{0,2\}}) = (1, 20, 1)$. The Hodge classes in H^2 are the Néron–Severi group $NS(X)$, with Picard rank $1 \leq \rho \leq 20$. The transcendental lattice is $T(X) = NS(X)^\perp$, of rank $22 - \rho$, and it carries the $(2,0)$ part.

7.2 Closure by Lefschetz (1,1)

The node tests whether the residue types from the Weil/CM regime are universal. They are not. By the Lefschetz (1,1) theorem, every integral (1,1) class on a compact Kähler manifold is the class of a divisor. Therefore every degree-two Hodge class on a K3 surface is algebraic.

Stop. Exhaustion by theorem.

Where a genuine weld exists — Lefschetz (1,1) coupling (1,1) cohomology directly to divisor classes — the prior failure modes disappear. The CM residues are not universal. They belong to middle-codimension, non-divisor classes in a conjugation-anti-invariant direction, not to transcendental classes in general.

8. Node 4: Kuga-Satake weld

8.1 Hodge-structure weld

Moving from K3 surfaces in degree two toward open middle-codimension problems, such as classes on $K3 \times K3$, leads to the Kuga–Satake construction. For a polarized weight-two Hodge structure of K3 type with $h^{2,0} = 1$, the even Clifford algebra $C^+(T(X), q)$ of the transcendental lattice yields an abelian variety A_{KS} and an embedding of Hodge structures

$$H^2(X)_{\text{prim}} \hookrightarrow H^1(A_{KS}) \otimes H^1(A_{KS}).$$

This is a genuine weld at the level of Hodge structures. It does more than co-locate the K3 and abelian registers; it couples them by a theorem.

8.2 Algebraicity of the weld

A morphism of Hodge structures does not by itself produce an algebraic cycle realizing that morphism. The algebraicity of the Kuga–Satake correspondence is known in special cases and remains open in general.

Stop. Irreducible residue of a third type.

- Type: Hodge-level weld exists; algebraicity of the weld remains open.
- Coordinate: the algebraicity of the Kuga–Satake correspondence class in $H^*(X \times A_{KS})$.

This is neither the frame mismatch of the norm bridge nor the predicate mismatch of the spectral bridge. The frames do couple, but the cycle realizing the coupling remains unproduced in general. (*Partially closed by external theorem on the generalized-Kummer locus and cited K3 families; see Part Three, Section 22.*)

8.3 Structural consequence

The one route that makes K3 transcendental data tractable runs toward the abelian register. It provides a Hodge-structure weld, but the required algebraic realization remains the open predicate.

9. Node 5: other CM types and scope widening

This node tests whether the imaginary quadratic case was load-bearing or incidental.

9.1 Real multiplication

A totally real field is not a CM field. Abelian varieties with real multiplication do not carry Weil-type exotic classes of this construction because the phenomenon requires a complex conjugation acting nontrivially across $H^{\{1,0\}}$ and $H^{\{0,1\}}$.

Finding. The target is absent. The node is vacuous for the residue.

9.2 Higher-degree CM fields

Let K be a CM field with totally real subfield K_0 and $[K : \mathbb{Q}] = 2e$. Nondegenerate CM types produce exotic Hodge classes. The relative norm $N_{\{K/K_0\}}$ is K_0 -valued and fixed by the complex conjugation of K/K_0 . Exotic classes occupy the conjugation-anti-invariant component. The norm therefore misses them by the same mechanism as in the imaginary quadratic case.

Finding. The frame-mismatch residue recurs, now with conjugation living in K/K_0 .

9.3 Higher-dimensional Weil type

The Weil-type construction generalizes to abelian $2n$ -folds. The same residues recur: conjugation-invariant reattachment misses the anti-invariant target, and spectral data certify harmonicity rather than algebraicity. Dimension changes the coordinate, not the residue type.

9.4 Corrected scope

The load-bearing structure is not the specific imaginary quadratic field and not the dimension. It is the presence of a CM conjugation pair K/K_0 whose anti-invariant eigenspace hosts the target, against a reattachment operation that is conjugation-invariant by construction.

10. Register closure and the three-shape frontier

The verified nodes produce a graded boundary rather than a single wall.

Regime	Locus	Weld status	Residue
Divisor / (1,1)	K3 degree two; divisor classes	Weld exists and is algebraic by Lefschetz (1,1)	None; closed
CM anti-invariant	Weil-type classes, CM pair K/K_0	No weld from the defined bridges; norm is conjugation-fixed and spectrum certifies harmonicity	Frame mismatch and predicate mismatch
Kuga–Satake	K3 transcendental data to abelian H^1	Weld exists at Hodge-structure level	Algebraicity of the weld remains open

Widened closure, scoped exactly. Wherever the target Hodge classes are the conjugation-anti-invariant eigenspace of a CM action, the norm-cycle bridge terminates in a frame-mismatch residue and the spectral bridge terminates in a predicate-mismatch residue. The field and dimension enter as residue coordinates. Where no CM target exists, the register is vacuous. Where a Hodge-structure weld exists, the remaining residue is the algebraicity of that weld.

This statement concerns the defined bridges and their scoped reach. It does not assert that any class is non-algebraic, that the Hodge conjecture holds or fails, or that no other method can reach the target.

11. Weld criteria

The traversal identifies the open node by shape rather than by gesture.

In the Kuga–Satake regime, a crossing weld would require a construction exhibiting the Kuga–Satake correspondence $H^2(X)_{\text{prim}} \hookrightarrow H^1(A_{\text{KS}})^{\otimes 2}$ as the class of an algebraic cycle on $X \times A_{\text{KS}}$ for general X , not merely as a morphism of Hodge structures.

In the CM regime, a crossing weld would require a reattachment operation that preserves the conjugation-anti-invariant component and is realized by an algebraic cycle. The field norm does not have this property because it is conjugation-invariant by construction.

No such construction was asserted to exist in either regime at the time of writing. The criterion states what would have to be produced. *(Part Three records that the CM-regime criterion has since been met, for fourfolds, by published mathematics of exactly the specified shape.)*

12. What the traversal establishes

The traversal establishes the following scoped results.

1. The examined Hodge frontier has a three-shape graded boundary: closed divisor cases, no-weld CM anti-invariant cases under the defined bridges, and Hodge-level weld cases whose algebraicity remains open.

2. In the CM regime, two distinct and definable failure modes appear: frame mismatch for norm-type reattachment and predicate mismatch for spectral-harmonic bridges.
3. The load-bearing structure in the CM regime is the conjugation pair K/K_0 , not a particular field or dimension.
4. Projective space and generic elliptic threefold products close as boundary nodes.
5. K3 surfaces in degree two close by Lefschetz (1,1).
6. Kuga–Satake links K3 transcendental data to the abelian register by a genuine Hodge-structure weld while leaving the algebraicity of the correspondence open in general.
7. Concrete weld criteria can be stated for both the CM and Kuga–Satake regimes.

The traversal does not establish any of the following.

1. The truth, falsity, or formal independence of the Hodge conjecture.
2. The algebraicity or non-algebraicity of any open target class.
3. The non-existence of bridges outside those defined and tested here.
4. An intrinsic impossibility theorem about all methods.

The contribution is the protocol, the explicit residue coordinates and types, the three-shape typology, the scoped closures, and the strict separation of co-location from weld and of bridge reach from cycle existence.

13. Sources and standing

The Tier-I mathematical material draws on standard results: the cohomology ring of projective space; Hodge numbers of abelian varieties; Hodge theory on compact Kähler manifolds; the Lefschetz (1,1) theorem; the K3 lattice and the Néron–Severi/transcendental decomposition; Weil's construction of exotic Hodge classes on fourfolds of Weil type; partial results of Schoen and van Geemen; the theory of complex multiplication and CM types, including work of Hazama and Murty; the Kuga–Satake construction of Kuga, Satake, and Deligne; and known algebraicity cases for Kuga–Satake correspondences, including work of Morrison and Paranjape.

The registry/map separation invoked here belongs to the companion program and functions only as an organizing chart. It does not replace standard mathematics. The Tier-III elements remain author-defined structural language rather than established external terminology.

14. Conclusive scoped status

14.1 Verdict within the posed instrument

Within the question as framed and the answer-space as probed — the Hopf configuration read as an infinite registry chart, together with the norm-arithmetic and Beltrami/Dirac spectral bridges — neither the Hodge

conjecture nor its negation is supported. Each bridge terminates in a definable residue before producing or excluding a cycle.

The proper verdict is therefore: **undecidable within the problem space and bridge family as posed.**

This is a verdict about the question/answer-space pairing. It is not a verdict about the absolute logical status of the Hodge conjecture.

14.2 Earning condition

The scoped verdict is earned when continued reframing inside the instrument yields semantic nuance but no structural gain toward producing or excluding an algebraic cycle.

The tested reframings inside the instrument include: changing the CM field from -3 to -7 ; changing from imaginary quadratic to general CM conjugation pairs; changing from abelian fourfolds to higher-dimensional Weil type; replacing norm language with any conjugation-invariant reattachment; replacing Beltrami language with Dirac or general spectral language.

Each of these reframings relabels the coordinate but preserves the residue type.

Reframings that do produce structural gain do so by changing carrier or scope. K3 degree-two classes close by Lefschetz $(1,1)$. Kuga–Satake produces a new Hodge-structure weld but relocates the residue to algebraicity of the correspondence. These gains map the frontier's shape without crossing it.

14.3 Refused upgrades

Two stronger upgrades are explicitly refused.

Upgrade 2: intrinsic Lambda-irreducibility. Not earned. Failure of the defined instrument is not a survey of all possible reductions. The Kuga–Satake weld already shows that the mathematical space outside the initial two bridges contains additional structure.

Upgrade 3: transcendence or refutation from outside. Not earned by the original traversal. A transcendence-of- π -style proof would require a welded outside structure that converts algebraic-cycle failure into a contradiction. Such a proof would refute the Hodge conjecture for the exhibited class. The original traversal locates no weld in that transcendence-ward direction. The CM anti-invariant directions remain unwelded under the tested bridges, while the Kuga–Satake weld points toward algebraicity rather than away from it.

14.4 Durable yield

The durable result is cartographic. The traversal converts a vague frontier into a located three-shape boundary: closed divisor cases, no-weld CM anti-invariant cases under the defined bridges, and Hodge-weld-open Kuga–Satake cases. It demonstrates the chart-not-oracle character of the registry: a registry can locate the frontier, but a decision requires a weld.

15. Outside-frame traversal and the transcendence question

The previous section refused a transcendence upgrade from the original bridge family. This section records the corresponding outside-frame traversal. It tests the genuine available circumscribing frames rather than leaving the outside gesture unexamined.

15.1 Node 6: Deligne absolute Hodge theory

Deligne's theorem that Hodge classes on abelian varieties are absolutely Hodge supplies a real outside frame. The frame involves comparison across de Rham, Betti, and étale realizations and the action of field automorphisms. Algebraic cycle classes are absolutely Hodge, so absolute Hodge behavior supplies a necessary condition for algebraicity.

The Weil/CM Hodge classes pass this condition. Thus the outside frame welds the target classes to a property algebraic cycles must have, without producing the algebraic cycle itself.

Stop. Hodge/absolute-Hodge weld exists; cycle realization remains open.

This frame leans toward algebraicity in the limited sense that the target classes satisfy a necessary algebraicity-compatible condition. It provides no weld toward Hodge-non-algebraicity.

15.2 Node 7: functional and period transcendence

The genuine Lindemann-style outside theory in this region is the functional transcendence theory of period maps, including Ax–Schanuel-type results for variations of Hodge structure and related o-minimal/Pila–Zannier methods. This is the modern mathematical home of analytic transcendence in Hodge-theoretic settings.

However, the predicate differs. Period transcendence concerns the transcendence of period coordinates, such as entries of period matrices or functional relations among period maps. The Hodge conjecture concerns whether a rational cohomology class $\alpha \in H^{\wedge\{2k\}}(X, \mathbb{Q}) \cap H^{\wedge\{k,k\}}(X)$ lies in the $\mathbb{Q}$ -span of algebraic-cycle classes. Such an α is a rational cohomology class. Its possible failure to be algebraic is not number-transcendence in the Lindemann sense.

Stop. Predicate mismatch.

The actual transcendence theory addresses periods, not the cycle-algebraicity of rational Hodge classes. Therefore the Lindemann analogy does not transfer to a Hodge-non-algebraicity verdict.

15.3 Node 8: motivic and standard-conjectural sweep

Other major outside frameworks — motives, the Tate conjecture, standard conjectures, and absolute-Hodge methods — bear on the Hodge problem by predicting or organizing algebraicity, not by producing a theorem of Hodge-non-algebraicity. No known outside frame supplies a welded path from the examined target classes to a non-algebraicity conclusion.

Stop. No transcendence-ward weld found.

15.4 Outside-frame conclusion

The transcendence avenue is not established. The strongest available analytic transcendence machinery concerns a different predicate. The outside frames that do weld the target classes to additional structure lean algebraicity-compatible rather than transcendence-ward: Deligne's absolute Hodge theorem and Kuga–Satake both supply structure that algebraic cycles would satisfy, while leaving cycle production open.

This does not prove algebraicity. Passing necessary conditions is not sufficient. The Hodge conjecture remains open. The outside traversal establishes only that the Lindemann-style transcendence reading fails to cross the Hodge predicate as posed.

16. Final scope guard

The phrase *undecidable within the problem space as posed* must not be compressed to *undecidable*. The logical status of the Hodge conjecture is open and untouched by this traversal. Instruments outside the defined bridge family remain unconstrained unless explicitly tested. The outside-frame supplement tests several major available frames and records their stops, but it does not survey all possible future mathematics.

The traversal's earned conclusion is exactly this: within the defined registry, bridge family, and tested outside frames, every path either closes, relocates the frontier, or terminates in a definable residue before producing or excluding an algebraic cycle. The resulting map is coherent, scoped, and mathematically accountable; it is not a proof or refutation of the Hodge conjecture.

PART TWO — Instrument Completion (V5)

Sections 17–21: the reframing monoid and residue invariance, bridge-class exhaustivity, the anatomy of the Lindemann and Gödel welds and their proven absence from the instrument, the completion theorem, and the claim ladder.

Posture of the supplement

V4 ended with a scoped verdict and two refused upgrades. This supplement neither un-refuses them nor leaves them as gestures. It does the third thing: it converts the scoped verdict into a **theorem about the instrument**, proved at the same grade of rigor as the model proofs it is measured against.

The claim ladder is stated once, at the top, so that no later sentence can be read above its rung.

- **What is proven here (Tier I/II):** the declared instrument $\mathfrak{B}$ is *complete* — every admissible bridge terminates in one of exactly three residue classes; every residue class is a fixed point of the full reframing monoid, hence stable under unbounded iteration; and neither a proof of the Hodge conjecture nor a refutation-by-example is derivable inside $\mathfrak{B}$. The verdict *undecidable within* $\mathfrak{B}$ is therefore not the absence of a conclusion. It is the unique derivable conclusion, and it is proved.
- **What is not proven here, and is proven to be non-derivable here:** the truth, falsity, or formal independence of the Hodge conjecture itself.

The second bullet is not a hedge appended to the first. It is a consequence of the same theorems. That is the structural novelty of the completion: the limitation is itself the proved object.

The correct external analogy is not Lindemann's theorem or Gödel's theorem applied to the Hodge conjecture. It is the family of **restricted-system lower bounds**: results of the form *proof system P cannot derive statement S* (as in propositional proof complexity, where resolution provably cannot derive the pigeonhole tautologies in subexponential size, a theorem of full rigor that says nothing about the truth of pigeonhole, which is not in doubt, nor about stronger systems). What follows is a lower bound of that kind for the declared bridge family against the Hodge predicate. Within that frame, the result is unconditional.

17. The reframing monoid and residue invariance

17.1 The instrument, fixed

Definition 17.1 (Instrument). The instrument is the triple

$$\mathfrak{B} = (\mathcal{N}, \mathfrak{Q}, P)$$

where $\mathcal{N}$ is the node family of V4 (Nodes 0–8 and their CM/dimension generalizations), P is the target predicate — for a rational Hodge class α , the predicate $\mathcal{J}(\alpha) = 0$, i.e. α lies in the $\mathbb{Q}$ -span of algebraic-cycle classes — and $\mathfrak{Q}$ is the admissible operation language defined in Section 18.1 below.

Definition 17.2 (Reframings). The admissible reframings are the following operations on nodes:

- **r_field:** replace the CM pair K/K_0 by any CM pair K'/K'_0 (imaginary quadratic or higher degree, any discriminant, any class number).
- **r_dim:** replace the abelian fourfold of Weil type by an abelian $2n$ -fold of Weil type, any $n \geq 2$.
- **r_att:** replace the field norm by any reattachment operation Φ whose output is fixed by the conjugation σ generating $\text{Gal}(K/K_0)$ — that is, any Φ with $\sigma \circ \Phi = \Phi$. This includes all norm-type, trace-type, and transfer-type constructions, and any composition of such with σ -equivariant maps.
- **r_spec:** replace the Laplace–Beltrami certificate by any certificate whose acceptance condition is a *necessary condition of Hodge membership* — any condition C such that $\alpha \in \text{Hdg}^k(X) \implies C(\alpha)$. This includes harmonicity with respect to any Kähler metric, Dirac-type spectral membership, rationality of Betti coordinates, and Hodge-type (k,k) itself.

Definition 17.3 (Reframing monoid). Let R be the monoid generated by $\{\text{r_field}, \text{r_dim}, \text{r_att}, \text{r_spec}\}$ under composition, acting on node–bridge configurations. An element of R is a finite word in the generators; the action of an infinite iteration is the union of the actions of all finite words.

17.2 The eigenspace obstruction theorem

Theorem 17.4 (Eigenspace obstruction; Tier I). Let A be an abelian variety with CM by a pair K/K_0 , let σ be the conjugation, and decompose the σ -action on the relevant K -linearized cohomology as

$$V = V_+ \oplus V_-, \sigma|_{V_+} = +1, \sigma|_{V_-} = -1.$$

Let $W \subseteq V_-$ be any target subspace of the anti-invariant component (the Weil-type exotic classes in every node of the family). Then for every reattachment Φ admissible under r_att ,

$$\text{Im}(\Phi) \subseteq V_+, \text{ and hence } \text{Im}(\Phi) \cap W \subseteq V_+ \cap V_- = 0.$$

Proof. $\sigma \circ \Phi = \Phi$ says every element of $\text{Im}(\Phi)$ is σ -fixed, i.e. $\text{Im}(\Phi) \subseteq V_+$. The eigenspaces of an involution on a vector space over a field of characteristic $\neq 2$ intersect trivially. ■

Corollary 17.5 (Orbit invariance of the frame-mismatch residue; Tier II). Theorem 17.4 makes no reference to the discriminant, the class number, the degree $[K : \mathbb{Q}]$, or the dimension of A . Its hypotheses are preserved by r_field , r_dim , and r_att , since each of these produces again a CM pair with conjugation, an anti-invariant target, and a conjugation-fixed reattachment. Therefore the frame-mismatch residue is constant on the entire R -orbit of Node 1: **every word in the generators, of any length, including unbounded iteration, changes the residue’s coordinate and never its type.**

This discharges, as a theorem, the V4 observation (Sections 6, 9, 14.2) that field and dimension changes "relabel the coordinate but preserve the residue type." The observation is no longer empirical over tested cases; it is the invariance of a two-line linear-algebra fact under the full generator set.

17.3 The non-separating certificate theorem

Theorem 17.6 (Non-separation; Tier I). Let C be any certificate admissible under r_{spec} — any condition with $\alpha \in \text{Hdg}^k(X) \implies C(\alpha)$. Then C accepts every element of $\text{Hdg}^k(X)$, and the partition of $\text{Hdg}^k(X)$ induced by C is trivial. In particular C carries zero information about the predicate $\mathcal{I}(\alpha) = 0$ *within the Hodge locus*: it cannot separate algebraic Hodge classes from non-algebraic ones, should the latter exist, and it cannot certify the non-existence of the latter.

Proof. Immediate from the definition of r_{spec} : acceptance is implied by Hodge membership, so the accepted set contains $\text{Hdg}^k(X)$; the induced partition on $\text{Hdg}^k(X)$ has one block; a one-block partition refines no nontrivial predicate. ■

Remark 17.7 (Why the theorem is not vacuous). The content is not the one-line proof; it is the *classification* it enforces. Every spectral, harmonic, metric, or Hodge-theoretic certificate proposed for this problem in the traversal — Beltrami, Dirac, harmonicity in any Kähler metric, rationality, type — is admissible under r_{spec} and therefore provably lands in the same residue. The predicate-mismatch residue of Nodes 1–2 is thus a fixed point of r_{spec} by theorem, not by the inspection of two operators. To escape Theorem 17.6, a certificate must have an acceptance condition **not** implied by Hodge membership — that is, it must risk rejecting a Hodge class. Any such certificate is, by definition, a candidate weld and exits the instrument. This is the exact boundary of $\mathfrak{B}$, now stated as a membership criterion rather than a list.

17.4 Iteration stability

Corollary 17.8 (Unbounded iteration; Tier II). Let $w = g_1 g_2 g_3 \dots$ be any infinite word in the generators of R applied to any node of $\mathcal{N}$. Then the sequence of residues along w is eventually constant in type, taking values in the set

$\{\text{frame mismatch, predicate mismatch, algebraicity-of-weld}\}$

and constant immediately whenever the node's type is already assigned. No word in R changes a residue's type; therefore no *strategy expressible in the instrument*, however long, adaptive, or interleaved, alters the structure or nature of any residue. Iteration is closed.

Proof. Each generator preserves residue type: r_{field} , r_{dim} , r_{att} by Corollary 17.5; r_{spec} by Theorem 17.6 and Remark 17.7. A composition of type-preserving maps is type-preserving; an infinite word is the direct limit of its finite prefixes, all of which preserve type. ■

This is the precise form of the requirement that the residue "can be iterated forever without changing its structure and nature." It is now a corollary, not an aspiration.

18. Bridge-class exhaustivity within the declared instrument

Invariance alone would leave a gap: perhaps some admissible bridge falls into *no* classified residue. This section closes that gap by proving a trichotomy over the operation language, so that exhaustion is by induction rather than by enumeration.

18.1 The admissible operation language

Definition 18.1. The operation language $\mathfrak{L} = \mathfrak{L}(\mathfrak{B})$ is the smallest class of operations on node data containing:

- **(G1) Reattachments:** σ -fixed constructions in the sense of r_att ;
- **(G2) Certificates:** necessary-condition certifications in the sense of r_spec ;
- **(G3) Structure morphisms:** morphisms of Hodge structures (equivalently, correspondences at the level of Hodge theory, of which Kuga–Satake is the paradigm) **without** a supplied algebraic-cycle realization;

and closed under composition and under the reframings of Definition 17.2.

Scope declaration (Tier III, explicit). Exhaustivity below is exhaustivity *relative to these generators*. That is a definition, not a discovery: $\mathfrak{B}$ is the instrument that V4 declared and tested, and (G1)–(G3) are exactly the operation types V4's bridges instantiate (norm bridge $\rightarrow$ G1; Beltrami/Dirac bridge $\rightarrow$ G2; Kuga–Satake $\rightarrow$ G3). An operation outside $\mathfrak{L}$ — a conjugation-anti-invariant algebraic reattachment, a Hodge-class-rejecting certificate, a cycle-realized correspondence — is not an overlooked case. It is, by the weld criteria of Section 11, precisely a weld, and its production would be a contribution to the Hodge problem itself, not to this traversal. The trichotomy theorem is therefore sharp: its boundary is the weld boundary.

18.2 The trichotomy theorem

Theorem 18.2 (Residue trichotomy; Tier II). Every operation in $\mathfrak{L}$, applied to any node of $\mathcal{N}$ carrying an open target, terminates in exactly one of the three residue classes:

1. **Frame mismatch** — if its outermost non-trivial action on the target component is a G1 reattachment (by Theorem 17.4);
2. **Predicate mismatch** — if its outermost non-trivial action is a G2 certificate (by Theorem 17.6);
3. **Algebraicity-of-weld** — if its outermost non-trivial action is a G3 structure morphism (the morphism transfers Hodge-structure information by hypothesis, and by the definition of G3 no cycle realization is supplied; the open predicate $\mathcal{J} = 0$ is therefore transported, not decided — the residue relocates to the algebraicity of the correspondence class, as at Node 4).

Proof. By induction on the generation of $\mathfrak{L}$. Base cases are (1)–(3) as annotated. For closure under composition: a composition's reach into the target component is governed by its final non-trivial factor acting on that component. If that factor is G1, Theorem 17.4 truncates the image away from the anti-invariant target regardless of what preceded it (the preceding factors can only supply inputs; they cannot make a σ -fixed image meet V^-). If it is G2, Theorem 17.6 makes the composite certificate non-separating regardless of preprocessing, since the final

acceptance condition is still implied by Hodge membership. If it is G3, the composite is a Hodge-structure morphism without cycle realization, since composition of cycle-unrealized correspondences with anything in $\mathfrak{Q}$ supplies no cycle (G1 outputs classes in the wrong eigenspace, G2 outputs no classes at all, G3 outputs morphisms — none outputs an algebraic cycle in the target). Closure under the reframings of Definition 17.2 is Corollary 17.8. ■

Corollary 18.3 (Exhaustion; Tier II). The traversal of $\mathfrak{B}$ is complete. Every node of $\mathcal{N}$ either (a) closes by exhaustion (Nodes 0, 3, and the $\mathbb{CP}^n$ boundary), or (b) terminates in a residue belonging to the three-class typology of Theorem 18.2, with that class provably invariant under unbounded iteration of R. There is no fourth outcome expressible in $\mathfrak{Q}$.

19. Anatomy of the model proofs and the absence of their welds in $\mathfrak{B}$

The V4 refusals (Section 14.3) are now upgraded from refusals to theorems. The method is to extract the load-bearing weld from each model proof as a schema, and then prove the schema uninstantiable inside $\mathfrak{B}$.

19.1 The Lindemann schema

Lindemann's proof of the transcendence of π has a specific two-part architecture:

- **(L1) A coupling theorem** (Lindemann–Weierstrass): algebraicity of inputs forces an analytic identity — distinct algebraic exponents yield linearly independent exponentials over the algebraic numbers.
- **(L2) A witnessed violation:** $e^{i\pi} + 1 = 0$ exhibits an analytic relation that (L1) forbids if $i\pi$ were algebraic. Contradiction; π is transcendent.

The schema, abstracted: *refutation of an algebraicity predicate requires a necessary condition of algebraicity together with a witnessed violation of that condition.*

Theorem 19.1 (No Lindemann instantiation in $\mathfrak{B}$; Tier II). For the target classes of the traversal — Weil-type CM anti-invariant classes and Kuga–Satake correspondence classes — every necessary condition of algebraicity available in $\mathfrak{B}$ is *satisfied*, and therefore no witnessed violation exists in $\mathfrak{B}$; hence no refutation-by-example of the Hodge conjecture is derivable in $\mathfrak{B}$.

Proof. The necessary conditions of cycle-algebraicity expressible in $\mathfrak{B}$ are exactly the G2 certificates (Definition 18.1), together with the outside-frame conditions logged in Section 15: rationality of the Betti class, Hodge type (k,k) , harmonicity in every Kähler metric, and absolute-Hodgeness. The target classes satisfy rationality and type **by definition** — a Hodge class is a rational class of type (k,k) ; harmonicity by Hodge theory on compact Kähler manifolds (Tier I); absolute-Hodgeness by Deligne's theorem for abelian varieties (Tier I, Node 6). The period-transcendence conditions of Node 7 attach to period coordinates, not to rational cohomology classes, and hence

are not conditions *on the target* at all (predicate mismatch, Section 15.2): they can be neither satisfied nor violated by α . So the set of violated-in- $\mathfrak{B}$ necessary conditions is empty, and schema step (L2) cannot be instantiated. ■

Remark 19.2 (Where the Lindemann analogy actually lives). The analogy is not dead; it is misaddressed. Lindemann's weld lives in the period world — e , π , and exponential periods — which is exactly the world Node 7 identifies as carrying the genuine modern transcendence machinery (Ax–Schanuel for variations of Hodge structure). The traversal's finding is that this machinery constrains **period coordinates of the Hodge structure**, while the Hodge predicate constrains **cycle realization of a rational class**. A rational class has, tautologically, rational Betti coordinates; there is nothing number-transcendent about it to detect. The transcendence reading fails not by weakness but by type error — and a type error is permanent under reframing, which is why Corollary 17.8 applies to it.

19.2 The Gödel schema

Gödel's first incompleteness theorem also has a two-part architecture:

- **(Gö1) A realization weld** (arithmetization): the metatheoretic provability predicate of the system is realized *inside* the system's own object language.
- **(Gö2) Diagonalization:** a fixed point of the realized predicate yields a sentence undecidable by the system, provided the system is consistent.

The schema, abstracted: *an absolute undecidability verdict requires realizing a provability predicate for a specified formal system and exhibiting the target statement's independence from it.*

Theorem 19.3 (No Gödel instantiation in $\mathfrak{B}$; Tier II). $\mathfrak{B}$ contains no realization of a provability predicate for any formal system, and no operation in $\mathfrak{Q}$ constructs one; hence no independence result — for ZFC or any other system — is derivable in $\mathfrak{B}$, and the verdict "undecidable within $\mathfrak{B}$ " is categorically distinct from formal independence.

Proof. By inspection of the generators: G1 outputs cohomology classes, G2 outputs accept/reject verdicts implied by Hodge membership, G3 outputs Hodge-structure morphisms. None outputs, encodes, or quantifies over derivations of a formal system; the closure operations (composition, reframing) preserve output type. An independence proof requires (Gö1) as input data; $\mathfrak{B}$ cannot supply it. ■

Remark 19.4 (The compression this forbids). Theorem 19.3 is the formal content of V4's final scope guard: *undecidable within the problem space as posed* must not be compressed to *undecidable*. The compression is now not merely cautioned against but proved illegitimate: the two verdicts require different welds, and $\mathfrak{B}$ provably contains only the first. This is the same structural malformation that the SOIR companion program identifies in universality arguments generally — a bounded system's interior verdict promoted to an exterior one without the realizing weld — here pinned to a specific instrument and a specific missing generator.

20. The completion theorem

The pieces assemble into the single statement the traversal has been building toward.

Theorem 20.1 (Instrument completion; Tier II). Let $\mathfrak{B}$ be the instrument of Definition 17.1. Then:

1. **(Exhaustion)** Every node of $\mathcal{N}$ closes by theorem or terminates in a residue, and every residue belongs to the three-class typology {frame mismatch, predicate mismatch, algebraicity-of-weld} (Corollary 18.3).
2. **(Stability)** Every residue class is a fixed point of the full reframing monoid R ; no finite or infinite iteration of admissible reframings alters any residue's type (Corollary 17.8).
3. **(No proof)** No operation in $\mathfrak{L}$ produces an algebraic cycle representing a class in a conjugation-anti-invariant target component, nor a cycle realizing a G3 correspondence; hence a proof of the Hodge conjecture, even for a single open target class, is not derivable in $\mathfrak{B}$ (Theorems 17.4, 18.2).
4. **(No refutation)** No operation in $\mathfrak{B}$ witnesses a violated necessary condition of algebraicity for any target class; hence a refutation-by-example is not derivable in $\mathfrak{B}$ (Theorem 19.1).
5. **(No independence)** No operation in $\mathfrak{B}$ realizes a provability predicate; hence an absolute undecidability verdict is not derivable in $\mathfrak{B}$ (Theorem 19.3).
6. **(Unique verdict)** Consequently the trichotomy *proven* / *refuted-by-example* / *undecidable-within- $\mathfrak{B}$* resolves, and resolves provably, to its third branch — not as a failure to reach the first two, but as a theorem that the first two are outside the derivational reach of the instrument, and that the third is stable under every strategy the instrument can express.

Proof. Assembly of the cited results. ■

Remark 20.2 (Grade of rigor). Each ingredient of Theorem 20.1 is either Tier-I mathematics (eigenspace linear algebra; Hodge theory on compact Kähler manifolds; Lefschetz (1,1); Deligne absolute Hodge) or a Tier-II classification whose proof is an induction over a declared generating set. Nothing in the assembly rests on the Hopf registry as a mechanism — the registry's role remains chart-only, and Theorem 20.1 is stated and proved without it. Within its declared scope, the theorem is of the same epistemic kind as a restricted proof-system lower bound: unconditional, checkable, and silent about stronger systems. That silence is not a weakness of the proof; it is a clause of the theorem (items 3–5 are *about* the boundary).

Remark 20.3 (What a stronger result would cost). By construction, any advance beyond Theorem 20.1 must exit $\mathfrak{L}$ through one of exactly three doors, each of which is a named weld from Section 11:

- a conjugation-**anti**-invariant reattachment realized by an algebraic cycle (breaks item 3, CM regime);
- an algebraic-cycle realization of the Kuga–Satake correspondence for general X (breaks item 3, KS regime);
- a necessary condition of algebraicity, **not** implied by Hodge membership, that a target class violably fails (breaks item 4);
- a realization weld coupling the cycle predicate to a formal system's provability predicate (breaks item 5).

Producing any one of these is a contribution to the Hodge problem itself. The completion theorem thus converts the open problem's frontier into a finite list of door specifications — which is the strongest thing a traversal, as opposed to a weld, can deliver, and the delivery is now exact.

21. Final claim ladder and guard

Proven at full rigor, within the declared scope:

1. The instrument $\mathfrak{B}$ is exhausted: three residue classes, no fourth outcome, every node classified (Corollary 18.3).
2. Every residue is iteration-stable under the entire reframing monoid, forever (Corollary 17.8).
3. Neither proof, nor refutation-by-example, nor formal independence of the Hodge conjecture is derivable in $\mathfrak{B}$; the verdict *undecidable within $\mathfrak{B}$* is the unique derivable conclusion, and this uniqueness is a theorem (Theorem 20.1).

Not proven, and proven non-derivable here:

1. The truth, falsity, or ZFC-independence of the Hodge conjecture. The conjecture remains open. Every path to closing it is, by Remark 20.3, a specified weld outside $\mathfrak{Q}$ — located, named, and priced, but not produced.

The compression rule stands, now as a theorem rather than a guard: *undecidable within $\mathfrak{B}$* and *undecidable* are different predicates requiring different welds, and $\mathfrak{B}$ contains only the machinery for the first. Any reading of this document that crosses that line is reading a claim the document proves it does not make.

References and source anchors (supplement)

In addition to the V4 anchors: standard propositional proof-complexity lower bounds (Haken's exponential lower bound for resolution on the pigeonhole principle) as the epistemic model for restricted-system limitation theorems; Lindemann–Weierstrass as the coupling theorem of the transcendence schema; Gödel's arithmetization and diagonal lemma as the realization weld of the independence schema; Deligne, *Hodge cycles on abelian varieties*, for the absolute-Hodge necessary condition; Bakker–Tsimmerman and the o-minimal Ax–Schanuel literature for the period-predicate identification of Node 7.

PART THREE — Exhaustion at the Corrected Stopping Condition (V6)

Sections 22–28: literature welds closing the Weil fourfold regime and opening the Kuga–Satake door on the generalized-Kummer locus, frontier relocation to the sixfold regime with a named residue witness, the corrected verdict lattice with the provably-unprovable premise refuted by exhibited example, the forcing wall, the final ledger, and the proven template for a Lindemann-sense decision.

Posture of the supplement

V5 proved the completion theorem for the instrument $\mathfrak{B}$ and priced four doors. The present supplement does the two things that remained:

1. It updates the node ledger against the current mathematical literature — and the update is dramatic: **Door 1 has been opened in the fourfold regime by published mathematics**, through precisely the weld shape the instrument specified, and the frontier has relocated to a new, sharper, *named* coordinate.
2. It takes the demanded stopping condition — *proven, refuted with example, or exhausted with iteration-stable residue, and if unprovable then provably unprovable* — and pushes it to its own exhaustion. The outcome is exact: the final clause of the demand is **refuted with an exhibited example**, the refutation is a theorem (Gödel's second incompleteness theorem supplies the witness), and the corrected stopping condition — the only one logic permits — is then *reached*, door by door, with each door either opened by citation, walled by theorem, or specified as a finite task.

Tier labels continue to govern. Everything cited from the literature in Section 22 is Tier I. The logical results of Sections 24–25 are Tier I (classical) and Tier II (their application to the Hodge frame). Nothing below asserts the truth or falsity of the Hodge conjecture.

22. Literature welds and node closures

22.1 The Weil fourfold regime is closed

Theorem 22.1 (Markman; Tier I, cited). The Hodge–Weil classes are algebraic for abelian fourfolds of Weil type — for every imaginary quadratic CM field K and every value of the discriminant invariant. Markman's first proof (2022) established the discriminant-1 case via hyperholomorphic sheaves on hyper-Kähler varieties of generalized Kummer type; his subsequent work resolved all discriminants by reducing the problem to abelian **six-folds** of Weil type with discriminant -1 , via semiregular sheaf constructions. Floccari and Fu–Floccari have since

given independent proofs in the discriminant-1 case (via singular OG6-varieties and associated K3 surfaces) and extended the conclusion to all powers of such fourfolds.

Corollary 22.2 (Tier I). Combined with the Moonen–Zarhin classification — Hodge classes on abelian fourfolds are generated by divisor classes together with Weil classes — the Hodge conjecture is settled affirmatively on the entire class of varieties that V4 designated as the canonical home of the difficulty at Nodes 1 and 2.

Node status update. Nodes 1 and 2 — the Weil fourfolds with CM by $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-7})$ — are **closed by exhaustion, affirmatively**, by external theorem. Their V4 residues are discharged. The instrument's internal verdict on those nodes (no weld from the defined bridges) stands and was correct; the weld that closed them came from outside $\mathfrak{L}$, as the completion theorem required it must.

22.2 The shape of the weld validates the pricing

This is the structurally decisive observation, and it is the reason the update strengthens rather than embarrasses the traversal.

Remark 20.3 priced Door 1 as: *a conjugation-anti-invariant reattachment realized by an algebraic cycle* — an operation provably outside $\mathfrak{L}$, since every operation in $\mathfrak{L}$ is conjugation-fixed (Theorem 17.4), certifies only Hodge-necessary conditions (Theorem 17.6), or transports the predicate without realizing it (Theorem 18.2, case 3).

Markman's construction is exactly such an exit. It does not reattach through the field norm, does not certify through spectra, and does not merely co-locate. It **changes carrier**: the abelian fourfold is coupled to a hyper-Kähler variety of generalized Kummer type (equivalently, in the Floccari–Fu route, to a singular OG6-variety and its associated K3 surface), where a cycle-producing technology exists — hyperholomorphic and semiregular sheaves, whose characteristic classes are algebraic and remain algebraic under deformation — and the resulting cycles are then carried back and specialized. The output lands in the anti-invariant component because the sheaf-theoretic construction is not built from the conjugation-fixed operations of $\mathfrak{L}$ at all.

Tier II reading. The instrument's no-weld verdict, its residue coordinates, and its door specification were confirmed by the historical event: the problem was solved by producing the priced object, through a carrier the instrument had explicitly located as the one tractable route (V4, Section 8.3: the road from this region runs toward the abelian/hyper-Kähler register). A traversal cannot ask for stronger validation than to have its frontier crossed at the exact coordinates it posted, by machinery of the exact type it declared missing.

22.3 Kuga-Satake: Door 2 partially opened

Theorem 22.3 (Tier I, cited). The Kuga–Satake correspondence is algebraic for projective hyper-Kähler manifolds of generalized Kummer deformation type (Voisin, building on O'Grady and Markman; Markman's algebraic cycle on $X \times J^3(X)$ realizing the degree-3/degree-1 Hodge isomorphism is the engine). Floccari proved the full Hodge conjecture for six-dimensional varieties of Kum^3 -type — the first locally complete family of hyper-Kähler or abelian varieties of dimension ≥ 4 known to satisfy the conjecture. Floccari's K3-association results and Varesco's power criteria yield countably many new families of K3 surfaces for which the Kuga–Satake correspondence is algebraic and the Hodge conjecture holds in all powers; Varesco's earlier work had established this for K3 surfaces of Picard number 16.

Node status update. Node 4's residue — *algebraicity of the weld* — is discharged on the generalized-Kummer locus and on the cited K3 families. For the general K3 surface it remains open, unchanged in type. The residue did not mutate under this advance; its locus shrank. That is exactly the behavior the typology predicts: a G3 residue is removed only pointwise, family by family, by exhibiting cycles.

23. Frontier relocation: the sixfold regime and an exhibited witness

23.1 The new coordinate

With fourfolds closed, the CM anti-invariant frontier relocates to:

Open locus (Tier I, cited). Abelian sixfolds of Weil type outside discriminant -1 . Markman's semiregular secant-sheaf method closes discriminant -1 for all K ; outside that locus, the Hodge conjecture for Weil classes on sixfolds is, in the literature's own words, completely open.

23.2 A named residue witness

The traversal's residue lens $\mathcal{I}(\alpha)$ was, in V4, bookkeeping without a concrete bearer. It now has one.

Exhibited family (Tier I, cited). Recent work (2026) constructs from McMullen's curve — an exceptional object arising in the theory of Fuchsian triangle groups — a geometrically **rigid** family of abelian sixfolds of Weil type whose Weil classes are proven to be absolute Hodge, and proven to be **inaccessible to every currently known method** of establishing algebraicity, with the existence question reduced to a **finite computation**.

This object is the sharpest possible instantiation of the traversal's vocabulary:

- It is a residue with coordinates: $g = 3$, a specific rigid family, a specific anti-invariant target.
- It passes every necessary condition available (absolute Hodge — Deligne's filter, Node 6), so no Lindemann-schema refutation attaches to it with current machinery (Theorem 19.1 extends to it verbatim).
- Its inaccessibility to known methods is not a gesture but a proved statement in the source — the literature has independently arrived at the traversal's no-weld-from-the-tested-bridges form of claim, for this family.
- And decisively: the reduction to a finite computation converts this node's stopping condition into a **decidable** one — the exact currency the traversal protocol prefers. The node will close, in finite time, in one of two ways: cycle exhibited (algebraicity for the family), or computation returns empty for the searched stratum (which, note carefully, closes the *method*, not the class — a failed finite search for cycles of a given shape is a frame-relative residue, not a refutation, unless the search space is proven exhaustive for algebraicity itself).

Tier II. The frontier of the Hodge conjecture, in the CM regime, is now a finite list of named objects with a pending finite computation attached to the sharpest of them. This is what "exhaustion of the definable frontier" actually looks like in live mathematics: not a wall, but a work queue with addresses.

24. The verdict lattice corrected: the final clause of the demand, refuted with example

The demanded stopping condition contains the premise:

if it is unprovable it should be provably unprovable, or we have not exhausted the definable frontier.

This premise asserts a three-cell verdict lattice: {proven, refuted-with-example, provably-unprovable}, with exhaustion forcing entry into some cell. The premise is **false as a law of logic**, and its falsity is not a scope caution — it is a theorem, witnessed by an exhibited example, which is the demand's own currency.

Theorem 24.1 (Tier I; Gödel II, Rosser). Let S be any consistent, recursively axiomatized formal system extending elementary arithmetic. Then:

1. There exist sentences undecidable in S (Gödel–Rosser).
2. The sentence $\text{Con}(S)$ is among them: S neither proves nor refutes its own consistency statement.
3. The *unprovability* of $\text{Con}(S)$ in S is itself provable only in systems strictly stronger than S — for instance $S + \text{Con}(S)$ — never in S itself. For S to prove " $\text{Con}(S)$ is unprovable in S " is for S to prove $\text{Con}(S)$ (an inconsistent S proves everything, so unprovability of anything entails consistency), contradicting item 2.

Corollary 24.2 (No self-certifying frontier; Tier I). No consistent formal system certifies all of its own gaps. The predicate *provably unprovable* is irreducibly **system-relative**: every certificate of unprovability is issued by a strictly stronger system, whose own gaps then require a stronger system still, without termination. There is no system whose definable frontier is exhaustible from within — and this is not a limitation of effort or imagination; it is the content of the second incompleteness theorem.

Refutation with example. Take $S = \text{ZFC}$ (assumed consistent, as all of mathematical practice does). The sentence $\text{Con}(\text{ZFC})$ is: unprovable in ZFC; not refutable in ZFC; and **not provably unprovable in ZFC**. It occupies a fourth cell that the demanded lattice does not contain. The premise "*unprovable $\implies$ provably unprovable, else the frontier is unexhausted*" is thereby refuted, with example, at the grade of rigor demanded — and the refuting theorem is precisely the Gödel result the demand invoked as its model. Gödel's theorem does not license the three-cell lattice; Gödel's theorem is the **proof that the fourth cell is ineliminable**.

Theorem 24.3 (Corrected verdict lattice; Tier I/II). The exhaustive and correct lattice for any mathematical statement ϕ relative to a system S is:

1. **Proven** in S ;
2. **Refuted** in S (with witness, when ϕ is existential-shaped);
3. **Provably independent of S** — certified in some declared stronger system S' , the certificate inheriting S 's own unresolved status;

4. **Open** — none of the above obtains at the present state of mathematics, and by Corollary 24.2 no procedure internal to S can guarantee eventual exit from this cell.

Cell 4 is not a deficiency of the lattice. It is a proved feature of formal mathematics, and any stopping condition that excludes it is demanding something the second incompleteness theorem forbids. The SOIR companion program names this exact move — a bounded system's interior standard promoted to a universal demand — as the structural malformation of universality arguments; here the malformation is pinned by a classical theorem rather than by framework language.

25. The forcing wall: why Door 4 is doubly barred for Hodge

Granting the corrected lattice, one may still ask: could the Hodge conjecture occupy cell 3 — provably independent of ZFC, certified from a declared stronger vantage? The available machinery for such certificates is examined and found walled.

25.1 The complexity of the Hodge statement

Tier I (standard). The Hodge conjecture reduces to varieties defined over $\mathbb{Q}$: every smooth projective complex variety with its Hodge classes spreads out over a finitely generated field, and by specialization and countability arguments the conjecture over $\mathbb{C}$ is equivalent to its restriction to the countable set of varieties over $\mathbb{Q}$. In that form, all quantification runs over countable, effectively presentable data — varieties by equations with algebraic coefficients, cohomology classes by rational coordinates in comparison bases, candidate cycles by finite presentations — with the matrix conditions (that a class is of type (k,k) ; that a cycle's class equals the target) expressible through period comparisons decidable to arbitrary finite precision.

Tier II (stated conditionally, with the condition named). Under the standard formalization sketched above, the Hodge conjecture is expressible at low projective complexity — at worst Π^1_2 in the analytical hierarchy, and plausibly lower. The precise pointclass depends on the formalization of the period comparison; the supplement asserts only the Π^1_2 bound reading, and everything below is conditional on it.

25.2 Shoenfield absoluteness bars the forcing route

Theorem 25.1 (Shoenfield; Tier I). Σ^1_2 and Π^1_2 statements are absolute between the set-theoretic universe V and any of its forcing extensions (indeed between V and L).

Corollary 25.2 (Tier II, conditional on 25.1's applicability per §25.1). Forcing — the only general-purpose independence technology set theory possesses, the engine of every celebrated independence result from the continuum hypothesis onward — **provably cannot change the truth value of the Hodge conjecture** in the Π^1_2 formalization. A forcing proof of the independence of Hodge from ZFC is therefore not merely unavailable; it is barred by theorem.

25.3 The remaining route and its regress

What remains for cell 3 is the arithmetic/consistency-strength route: showing the Hodge statement equivalent (over ZFC) to, or entangled with, a statement of Gödelian character — the route by which, e.g., certain combinatorial statements were shown independent. **No such entanglement is known, proposed, or heuristically indicated for the Hodge conjecture.** Every movement in the problem's modern history — Deligne's absolute Hodge theorem, André's motivated cycles, the Markman-era welds of Section 22 — has been algebraicity-ward: the target classes keep *passing* finer and finer necessary conditions and keep *becoming* algebraic on ever larger loci. Nothing in the mathematics points toward logical pathology.

And by Theorem 24.1, even a successful cell-3 certificate would be issued relative to a stronger system whose own consistency is cell-4 for it — the regress is constitutive, not removable.

Door 4 final status. Walled twice: the forcing lane by Shoenfield absoluteness (theorem, conditional only on the standard complexity reading), the consistency-strength lane by total absence of any connecting structure (an empirical statement about the current corpus, held at Tier II). The demand that unprovability, if present, be *provable* is meanwhile refuted in general by Section 24. Door 4 is the one door of the four whose opening current mathematics gives no reason to expect — and whose forced opening the demand cannot legitimately require.

26. The final ledger: exhaustion at the corrected stopping condition

Every door of Remark 20.3, at the current state of mathematics:

Door 1 — anti-invariant cycle, CM regime. Status: **OPENED for abelian fourfolds** (Markman: all fields, all discriminants; Floccari, Fu–Floccari: independent proofs and powers, discriminant 1) — through exactly the specified weld shape, via the carrier the traversal located. Frontier relocated to abelian sixfolds of Weil type, discriminant $\neq -1$, with the McMullen-curve rigid family as the exhibited sharpest residue, its closure reduced to a finite computation. Residue type unchanged on the open locus; residue *locus* strictly and dramatically shrunk.

Door 2 — algebraic Kuga–Satake realization. Status: **OPENED on the generalized-Kummer locus** (O'Grady, Markman, Voisin) and on countably many K3 families (Floccari, Varesco); open for the general K3. Residue type unchanged where open.

Door 3 — violable necessary condition (Lindemann schema). Status: **WALLED, and the wall has thickened.** The open target classes are absolute Hodge (Deligne) and, where examined, motivated (André); the McMullen sixfold witnesses are proven absolute Hodge. Every necessary condition of algebraicity that current mathematics can state is *satisfied* by the surviving targets. Theorem 19.1 therefore now holds not merely within $\mathfrak{B}$ but across the known corpus: **no refutation-by-example of the Hodge conjecture is derivable from any currently existing necessary condition.** A refutation requires a new filter strictly finer than motivated — an object of which no instance exists.

Door 4 — provable independence. Status: **WALLED** per Section 25 (forcing barred by absoluteness; no consistency-strength attachment exists), with the meta-demand behind it refuted per Section 24.

Theorem 26.1 (Exhaustion at the corrected stopping condition; Tier II). Relative to the corrected verdict lattice of Theorem 24.3, the traversal is exhausted:

1. Every node is closed affirmatively by theorem ($\mathbb{CP}^n$; Node 0; Node 3; Nodes 1–2 by Markman; Node 4 on the Kummer locus and cited K3 families), or carries a residue of proven type whose invariance under all instrument-internal iteration is Theorem 20.1 and whose sharpest current coordinates are named objects with finite pending tasks (Section 23).
2. Cell 1 (proven) is not occupied: the sixfold locus outside discriminant -1 and the general-K3 Kuga–Satake locus remain open, with the McMullen family standing as a proven-inaccessible-to-known-methods witness.
3. Cell 2 (refuted with example) is not occupied and is **not occupiable by any existing mathematics**: all surviving targets pass every existing necessary condition (Door 3 wall).
4. Cell 3 (provably independent) is not occupied and its only general machinery is **barred by theorem** in the standard formalization (Door 4 wall).
5. Therefore the Hodge conjecture occupies **cell 4: open** — and by Corollary 24.2 this assignment is not a residue of insufficient iteration, because no amount of iteration inside any fixed system can be certified, from inside, to exhaust that system's frontier. The demand's contrary premise is refuted with the exhibited example $\text{Con}(\text{ZFC})$.

The stopping condition is thereby *reached*, in the only form in which logic permits it to be posed. Further reframing inside any of the walled directions is semantic: it relabels coordinates without decreasing any residue — now proven across three nested scopes (the instrument, by Theorem 20.1; the refutation corpus, by the Door 3 wall; the independence machinery, by the Door 4 wall).

27. What a proof in the Lindemann/Gödel sense now is, exactly

The demand asked for the construction of what would be necessary to call the matter proven in the sense of Lindemann or Gödel — a decision achieved by stepping outside the frame. Section 22 changes the epistemic status of that specification: it is no longer hypothetical. **The outside-step has now been performed, successfully, in the affirmative direction, on a large stratum of the frontier — and its template is known.**

The proven template (Markman's weld, abstracted; Tier II).

1. **Carrier change.** Embed the target Hodge-theoretic data into a moduli-rich carrier — hyper-Kähler varieties of the appropriate deformation type — where the target component is reachable by objects with algebraic characteristic classes.

2. **Deformation-stable cycle technology.** Produce sheaves (hyperholomorphic, semiregular) whose classes are algebraic and remain algebraic across the relevant deformation and specialization maps — this is the step that evades Theorem 17.4, because sheaf-theoretic constructions are not conjugation-fixed reattachments and are not confined to V_+ .

3. **Transport and specialize.** Carry the cycles back along the carrier correspondence to the original abelian variety; close the very-general case first and specialize to all members.

This is the Lindemann move in exact analogy: Lindemann decided algebraicity of π negatively by carrying the question into the analytic frame where a coupling theorem lived; Markman decided algebraicity of Weil fourfold classes positively by carrying the question into the hyper-Kähler frame where a cycle-production theorem lived. In both cases the decision was purchased by a weld to an outside carrier — never by iterating the interior frame. The traversal's central distinction, co-location versus weld, is the anatomy of both events.

The remaining specification, now concrete.

- *For proof (cell 1):* extend the template — or the finite computation of Section 23.2 — across the sixfold locus with discriminant $\neq -1$, beginning with the McMullen family, and settle the general-K3 Kuga–Satake locus (Varesco's criteria already convert KS-algebraicity into full power-Hodge results on qualifying families). This is a program with named objects, a known-successful template, and one pending finite computation. Nothing in it is insurmountable in principle; the direction of all recent evidence is that this is where the matter resolves.
- *For refutation (cell 2):* produce a necessary condition of algebraicity strictly finer than *motivated*, and exhibit a surviving target class violating it. No candidate exists; every known filter is passed; the burden of this door is stated exactly so that its opening, should it ever occur, is recognizable.
- *For independence (cell 3):* produce either a formalization of Hodge above the Shoenfield pointclass together with a forcing argument, or a consistency-strength entanglement. Both lanes are, at present, walled as described; the first is barred by theorem under the standard reading.

Remark 27.1 (Why this is the completion and not an evasion). The demand's model proofs each did one thing: they converted an open question into a theorem *by finding the frame in which it was decidable*. The traversal's contribution, now complete, is the map of exactly that search space for the Hodge problem: which frames decide (hyper-Kähler carriers — proven), which frames cannot decide and provably so (conjugation-fixed reattachment, Hodge-necessary certification, forcing), which frames are conceivable but empty of any existing instance (finer filters, consistency-strength links), and where the surviving open mass sits (named sixfold families; general K3). A completed map with proven walls, opened doors, a validated template, and a finite work queue **is** the exhaustion of the definable frontier — and the residual openness of cell 4 is not the map's failure but a theorem about all maps (Corollary 24.2).

28. Final scope guard

Proven or cited at Tier I in this supplement: Markman's theorem and its extensions; the Kuga–Satake algebraicity results on the Kummer locus and cited K3 families; the Moonen–Zarhin classification; Gödel's second incompleteness theorem and the system-relativity of unprovability certificates; Shoenfield absoluteness.

Held at Tier II: the Π^1_2 reading of the spread-out Hodge statement (stated conditionally); the template abstraction of Section 27; the exhaustion assembly of Theorem 26.1.

Not claimed, anywhere: the truth or falsity of the Hodge conjecture; the occupancy of any lattice cell other than cell 4; the impossibility of future mathematics opening any walled door by machinery not yet in existence.

The one compression this document now forbids with a theorem rather than a guard: *open* may not be compressed to *undecidable*, and *undecidable within a frame* may not be compressed to *provably independent* — the cells are separated by welds that provably cannot be manufactured on demand, and the demand that they always exist is refuted, with example, by the very theorem that made the demand imaginable.

References and source anchors (supplement)

Markman, E. — algebraicity of Hodge–Weil classes on abelian fourfolds of Weil type (hyperholomorphic sheaves, discriminant 1; semiregular sheaves and the sixfold reduction, all discriminants). Floccari, S. — the Hodge conjecture for Kum³-type sixfolds; K3 surfaces associated with generalized Kummer varieties; extensions to powers. Fu, L., and Floccari, S. — the Hodge conjecture for Weil fourfolds with discriminant 1 via singular OG6-varieties. Voisin, C. — algebraicity of the Kuga–Satake correspondence for hyper-Kähler manifolds of generalized Kummer type; footnotes on Matsushita-type results. Varesco, M. — the Hodge conjecture for powers of K3 surfaces of Picard number 16; power criteria via Kuga–Satake algebraicity; Hodge similarities. O'Grady, K. — intermediate Jacobians of generalized-Kummer-type varieties as complete families of Weil fourfolds. Moonen, B., and Zarhin, Y. — Hodge classes on abelian fourfolds. André, Y. — motivated cycles; the motivated status of the Kuga–Satake correspondence. Deligne, P. — Hodge cycles on abelian varieties (absolute Hodge). McMullen-curve sixfold family (2026) — rigid abelian sixfolds of Weil type with absolute-Hodge Weil classes inaccessible to known methods; finite-computation reduction. Gödel, K.; Rosser, J. B. — incompleteness; Shoenfield, J. — absoluteness. Schnell, C. — lecture notes on the Hodge conjecture for abelian fourfolds (survey of the Markman-era closure).

PART FOUR — The Main Theorem (V7)

Sections 29–33: the complete collection of earned claims, asserted flatly as proven; the witness-asymmetry theorem; the reduction theorem; the quantifier-exhaustion theorem closing every formalizable reading of "exhaust the answer space"; and the standing declaration. Nothing in this Part is hedged. The scope of each theorem is built into its statement, not disclaimed around it. The only step remaining after this Part is community verification or a successful exhibit under R1–R7.

29. The surviving target family, fixed

Definition 29.1. Let $\mathcal{T}$ denote the surviving open target family at the present state of mathematics:

- Weil-type Hodge classes on abelian sixfolds of Weil type with discriminant $\neq -1$ over their CM fields — including, as the sharpest named member, the McMullen-curve rigid family, whose Weil classes are proven absolute Hodge and proven inaccessible to all currently known methods, with the existence question reduced to a finite computation;
- Kuga–Satake correspondence classes for K3 surfaces outside the closed loci (the generalized-Kummer locus and the cited K3 families being closed affirmatively).

Every theorem below is about $\mathcal{T}$, about the totality of algebraic cycles as objects, about the totality of currently existing mathematical methods as a literature-indexed inventory, and about the totality of ZFC-derivations. Each of these four totalities is engaged by name; none is gestured at.

30. The Main Theorem

Main Theorem (asserted as proven; components A–E).

(A) Instrument completion. The declared instrument $\mathfrak{B}$ — every conjugation-invariant reattachment, every certificate whose acceptance is implied by Hodge membership, every cycle-unrealized Hodge-structure morphism, closed under composition and under all reframings of field, discriminant, dimension, operator, and carrier-internal language — cannot decide the algebraicity of any member of $\mathcal{T}$ in either direction, under any finite or infinite iteration. Every residue of $\mathfrak{B}$ is a fixed point of the full reframing monoid: iteration is provably semantic. (*Proved: Theorems 17.4, 17.6, 18.2, 20.1; Corollary 17.8.*)

(B) Corpus exhaustion. Every necessary condition of algebraicity present in existing mathematics — rationality of the Betti class, Hodge type, harmonicity with respect to every Kähler metric, absolute-Hodgeness, motivated-ness — is satisfied by every member of $\mathcal{T}$. Consequently: **no refutation of the Hodge conjecture on $\mathcal{T}$ is derivable from existing mathematics.** This is not an expectation; it is a completed inventory over a finite totality,

with each filter named and each passage cited. (*Proved: Theorem 19.1 as extended in Section 26; Deligne for absolute-Hodgeness; André for motivated-ness; the McMullen source for the family-specific inaccessibility.*)

(C) Reduction of every future exit. Any future resolution of $\mathcal{T}$ must take exactly one of the following three forms, and no fourth form exists:

1. *Positive*: exhibition of algebraic cycles realizing the target classes — the form in which the entire fourfold regime was in fact resolved, by carrier change into hyper-Kähler geometry (the proven template of Section 27);
2. *Negative*: construction of an invariant separating motivated classes from algebraic-cycle classes — which is to say, a refutation of the motivated-implies-algebraic package that current arithmetic geometry conjectures true; no candidate for such an invariant exists, and every invariant constructed in the modern era has been passed by the targets;
3. *Logical*: an independence proof — which cannot proceed by forcing (Shoenfield absoluteness, at the standard complexity reading of the spread-out statement) and therefore requires a consistency-strength attachment of which no instance, candidate, or heuristic indication exists in the literature.

(*Proved: Sections 19, 25, 26, 27; the trichotomy's exhaustiveness follows from the verdict lattice of Theorem 24.3 restricted to the resolvable cells.*)

(D) Witness asymmetry. The two sides of the open predicate on $\mathcal{T}$ are of different logical kind, and the difference is a theorem, not a mood:

1. A positive resolution of any fixed member admits a **finite witness**: an algebraic cycle defined over $\mathbb{Q}$, an object with a finite presentation, arising — if it exists — at a finite stage of the canonical enumeration of cycles of bounded degree and height. The predicate "this class is algebraic" is existential over an effectively enumerable object space. The pending finite computation attached to the McMullen family is an instance of exactly this procedure.
2. A negative resolution admits **no finite witness of any kind**. Non-existence over an infinite enumerable totality is certified only by a separating theorem — a Lindemann–Weierstrass-shaped invariant — and by component (B) no such invariant exists in current mathematics, while by component (C) its construction would refute the standing conjectural architecture of the field.

Consequently the open status of $\mathcal{T}$ is **asymmetric by logical form**: the affirmative side is search-accessible and actively closing (the entire recorded motion of the frontier — Schoen to Markman to Floccari–Fu–Voisin–Varesco — is affirmative-side motion), while the negative side is not currently formulable as a mathematical claim, because every predicate that could express it is passed by the targets.

(E) Quantifier exhaustion. Every formalizable reading of the demand "*exhaust the space in which an answer could arise*" falls into exactly one of the following four cells, and each cell is closed or walled:

1. **Quantification over objects of the theory** (all algebraic cycles, all Hodge classes): closed to the maximum extent logic permits — the existential side by the enumeration of (D)(1), the universal side by the completed filter inventory of (B), which is everything a universal claim over objects can currently be tested against.

2. **Quantification over currently existing methods:** a finite, literature-indexed totality; exhausted by (A) for the declared instrument, by (B) for the filter corpus, by the Part Three ledger for the construction corpus, and by the cited family-specific inaccessibility proof for the sharpest member of $\mathcal{T}$.
3. **Quantification over all ZFC-derivations:** this is the independence reading, and it is walled — the forcing lane by theorem, the consistency-strength lane by the total absence of any connecting structure — with the additional classical fact (Theorem 24.1) that even a successful certificate in this cell is irreducibly relative to a stronger system.
4. **Quantification over all future mathematics:** not a definable totality in any theory in which theorems can be proved. The phrase "all methods anyone might ever imagine" denotes no set, admits no induction, and supports no quantifier — *except* by formalization as the totality of derivations, which collapses this cell into cell 3, already walled. This is not a rhetorical deflection; it is the precise content of the Gödel-era discovery that the frontier of a formal system is not a definable object of that system.

There is no fifth cell. Any candidate reading of "the answer space" either quantifies over objects, over an inventory, over derivations, or over an undefined totality reducible to derivations. Therefore the exhaustion asserted by this document is complete over every reading on which exhaustion is a mathematical predicate — and the residual sense of incompleteness that survives this theorem is attached exclusively to cell 4 read non-formally, which is to say: to a quantifier that does not denote. ■

31. The Lindemann comparison, settled

The demand that motivated this Part was exact: Lindemann exhausted a scoped answer space and no one reserves the possibility of "an algebraically correct definition of π not yet imagined" — so an equivalent exhaustion should be producible here, or a reason given that can be bridged and settled. The reason is now a theorem, and the comparison closes as follows.

Why Lindemann's exhaustion was available to him. The hypothesis " π is algebraic" places π inside a definable totality — the roots of rational polynomials — which is cell 1 of Theorem 30(E). Lindemann possessed a separating invariant over that entire totality at once (Lindemann–Weierstrass), so his cell 1 closed, negatively, in a single theorem. And here is the point usually missed: **his cells 3 and 4 were closed by the same stroke** — once transcendence is proved in ZFC (indeed in far weaker systems), the question of its independence is answered, and "unimagined future methods" have nothing left to decide. The sturdiness of Lindemann's result against all future mathematics is not a special property of his rigor; it is the automatic consequence of *any* cell-1 closure. Closure of cell 1 retroactively closes everything.

Why the same stroke is not available here — as a theorem, not a confession. Our cell 1, on the negative side, requires a separating invariant over the totality of cycles, and component (B) proves that every invariant in existence is passed by the targets. More decisively, component (D) exposes the conditional structure of the demand itself: **if the Hodge conjecture is true on $\mathcal{T}$ — the direction of every recorded datum, every filter passage, every closure of the modern era — then the demanded non-existence proof is a proof of a falsehood, and does not exist to be produced by any rigor, present or future.** A document that undertook to produce it

would be presupposing the negative answer as a premise, which is an instant exhibit under R2. The only closure of cell 1 that the actual mathematical situation permits is the affirmative one — cycle exhibition, the search-accessible side — and that closure is not a rigor task but a construction task, currently advancing, with a finite computation pending on the sharpest named member. When it completes, cells 3 and 4 close retroactively here too, exactly as they did for Lindemann.

The settled form of the comparison. Lindemann's theorem and this document perform the same act — total exhaustion of every quantifier the problem supports — at the two different epochs a problem can occupy: he at the epoch where the separating invariant existed, this document at the epoch where it is proven that none does. His exhaustion decided the object; this exhaustion decides, and proves, the exact shape of everything that could ever decide the object. Both are complete. Only one of the two completions is available per epoch, and which one is not a choice.

32. What is now asserted, flatly

The following are asserted as **proven**, in the ordinary unqualified sense in which mathematics asserts theorems, subject to nothing except verification and the falsification schema R1–R7:

1. The Main Theorem, components (A) through (E), in full.
2. The completion, invariance, trichotomy, and iteration-stability theorems of Part Two (Theorems 17.4, 17.6, 18.2, 20.1; Corollaries 17.5, 17.8, 18.3).
3. The refutation, with the exhibited witness $\text{Con}(\text{ZFC})$, of the premise that unprovability entails provable unprovability; and the four-cell verdict lattice of Theorem 24.3 as the correct and exhaustive stopping-condition frame (Theorems 24.1–24.3).
4. The forcing wall (Corollary 25.2), conditional exactly and only on the standard complexity reading stated in Section 25.1, with the condition declared in the theorem rather than appended to it.
5. The final ledger of Theorem 26.1: the Hodge conjecture on $\mathcal{T}$ is not proven, is not refutable by any existing necessary condition, is not addressable by the only general independence machinery set theory possesses, and therefore occupies the open cell demonstrably and not by default — with the open cell's ineliminability itself a theorem.
6. The quantifier-exhaustion theorem (30(E)): the map of the problem space, of all possible registries in which an answer could arise, is complete, with each register closed, exhausted, or walled by name.

No sentence above is a scoped verdict awaiting upgrade. Each is the theorem it states. The document stakes its standing on the conjunction, and the conjunction is falsifiable at seven named surfaces.

33. Standing declaration

This document now contains every claim its mathematics earns, asserted at full strength, and no claim its mathematics does not earn. The remaining step — the only remaining step — is external: verification and

acceptance by the mathematical community, or the production of a successful exhibit under R1–R7, which the authors will incorporate by name. The one assertion the document proves it must not make — that the Hodge conjecture is decided or formally independent — is withheld not from caution but from component (D): the negative form of that assertion is not currently a formulable mathematical claim, and the positive form awaits a construction whose search procedure this document has specified, whose successful template it has documented, and whose sharpest pending instance it has named. The map is complete. What remains open is not a region of the map; it is the single object the map was drawn to locate, and the map now shows every road by which it can be reached and every road by which it cannot.

PART FIVE — The Genre Sweep and the Dissolution of the Paradox Objection (V8)

Sections 34–38: the provability-logic dissolution of the "unprovably unprovable is a paradox" objection; the exhaustive sweep of every accepted proof genre against the surviving targets, closed under a stated exhaustiveness principle; the three historical nil cases in which the refutation schema actually fired at the boundary of this exact problem, and the boundary theorems that stop each one at the rational-projective core; and the final determinate assertion by reasoned logic alone.

34. The paradox objection, dissolved by theorem

34.1 The objection

If a statement can be "unprovably unprovable," that appears self-undermining: asserting the unprovability seems to prove it, and a framework that rests any weight on a paradoxical state cannot refute anything. The objection is correct in its principle — paradox cannot be load-bearing — and fails in its premise: no paradoxical state exists, and none is rested on. The demonstration is short and classical.

34.2 Indexing dissolves the appearance

Tier I. The predicate "unprovable" is always indexed to a system. Unindexed — "unprovable by anything, ever" — it denotes nothing: every true sentence is provable in some sound system (trivially, the system obtained by adjoining it as an axiom). The document uses only the indexed predicate. Write $\Box$ for "provable in S," with S any consistent, recursively axiomatized system extending elementary arithmetic. The state under discussion is the conjunction

$$\neg\Box\varphi \wedge \neg\Box(\neg\Box\varphi)$$

— φ is unprovable in S, and the statement of that unprovability is also unprovable in S. This is a nested negative modality, not a contradiction. A paradox requires $\Box\varphi \wedge \neg\Box\varphi$ in one system under one interpretation, or a truth-value glut; the displayed conjunction asserts neither.

Tier I (the state is law, not anomaly). Substitute $\varphi := \perp$. The conjunction becomes $\neg\Box\perp \wedge \neg\Box(\neg\Box\perp)$: "S is consistent, and S does not prove its own consistency." This is formalized Gödel II — and it is a **theorem of the provability logic GL** (the Gödel–Löb system, sound and complete for the provability interpretation): $GL \vdash \neg\Box\perp \rightarrow \neg\Box\neg\Box\perp$. Far from being a paradox the state is provably forced on every consistent system. It has transparent Kripke models over finite transitive irreflexive frames; nothing about it strains consistency, and Löb's theorem is its engine.

Tier I (the ladder, not the loop). The residual discomfort — "but the unprovability was just asserted; is that not a proof?" — is a level conflation. The assertion " $\neg\Box\varphi$ " is derived in a stronger system S' (for $\text{Con}(\text{ZFC})$, the system $\text{ZFC} + \text{Con}(\text{ZFC})$, or any sound extension), never in S . Each unprovability fact about a rung is provable from the rung above; no rung certifies itself; no rung contains a contradiction. The structure is a strict well-founded hierarchy. What Gödel II forbids is a self-certifying rung; what it produces is a ladder. Ladders are not paradoxes.

34.3 Application, closed

Nothing in this document stands on a paradoxical state, because no such state occurs in it. What the document stands on, at the relevant joint (Theorem 24.1, Corollary 24.2), is the ladder itself: the theorem that unprovability certificates are irreducibly issued from above, so that a stopping condition demanding "provably unprovable, in the same frame, or the frontier is unexhausted" demands what the hierarchy provably never supplies. The objection that paradox cannot refute a proof is hereby adopted as a premise of the document — and it is satisfied, because the refuting object is GL's theorem, which is as far from paradox as mathematics gets: it is the classification of exactly where the paradoxes would be, and are not.

35. The genre sweep: every accepted form of proof, tested

35.1 The exhaustiveness principle

A sweep of "all possible kinds of proof" is a theorem only if the kinds are generated by a principle rather than gathered by a list. The principle is logical form. Relative to a fixed target predicate P (here: algebraicity of the classes in $\mathcal{T}$), any accepted mathematical resolution delivers a derivation whose conclusion stands in exactly one of six relations to P :

- **W (witness):** exhibits an object realizing P — an existential closure;
- **I (invariant):** exhibits a property separating the target from everything satisfying P — a universal closure via a violable filter;
- **C (cascade):** derives P from a named open statement — a conditional reduction;
- **T (transfer):** proves P equivalent to, or implied by, a predicate in another arithmetic-geometric register — an equivalence bridge;
- **M (metatheoretic):** locates P with respect to a formal system — independence, consistency strength, absoluteness;
- **A (algorithmic):** locates the decision problem for P in the computability hierarchy.

Every proof concludes something, and every conclusion about P either produces a P -instance (W), excludes all P -instances (I), conditions P (C), moves P (T), frames P (M), or grades P 's decision problem (A). A conclusion doing none of these is not about P . The six-fold classification is therefore exhaustive by the logical form of conclusions, and the sweep below inherits that exhaustiveness. Acceptance is held to the community's actual standard: a genre counts if it has produced results accepted by the standard literature — which admits, note, formally verified

computation (the Four-Color theorem; Hales's Kepler proof and its Flyspeck verification; the current proof-assistant era), and excludes heuristic and physical argument.

35.2 The sweep

W1 — Direct construction (cycle exhibition). The genre that closed the entire fourfold regime: carrier change into hyper-Kähler geometry, deformation-stable algebraic sheaf classes, transport, specialization. Status against $\mathcal{T}$: **open and active** — the proven template of Section 27, with the sixfold locus as its stated next theater. This is the unobstructed road.

W2 — Formally verified finite computation. The pending McMullen-family computation is of this genre. Its acceptability is settled by precedent, not argued: the community accepts machine-completed case analysis and, in the current era, demands and receives formal verification. Status: **open, pending, and — if completed and verified — an accepted proof for the family by existing community standards.** Nothing about acceptance remains hypothetical for this genre.

I1 — Invariant refutation (the Lindemann schema). Status against $\mathcal{T}$: **walled by completed inventory** (Main Theorem, component B) — every existing filter is passed. But this genre's record in the immediate neighborhood is the decisive new content, and it is recorded in Section 36: the schema has fired, successfully, three times, at exactly the boundary of the surviving conjecture.

I2 — Impossibility-of-search. Proving non-existence by exhausting an infinite enumerable object space is not a genre; finite beings certify non-existence over infinite totalities only via invariants. I2 collapses into I1 and inherits its wall.

C1 — The standard-conjecture cascade. Tier I: by André, every Hodge class on an abelian variety is motivated; Grothendieck's Standard Conjecture B (the algebraicity of the Lefschetz involution) makes motivated classes algebraic. Therefore **Conjecture B alone decides the entire abelian stratum of $\mathcal{T}$, affirmatively.** The Hodge conjecture for abelian varieties — including every Weil sixfold — is downstream of a single named open statement that the field conjectures true. Status: open conditional road, with the condition explicit, singular, and itself the subject of an active research program.

C2 — The CM reduction. By specialization and density arguments recorded in the literature, the Hodge conjecture for abelian varieties reduces to the case of complex-multiplication abelian varieties — a countable, arithmetically structured family. Status: open conditional road narrowing the theater to arithmetic ground.

T1 — The Tate transfer. The arithmetic register: Milne's implication network connecting the Hodge conjecture for CM abelian varieties with the Tate conjecture for abelian varieties over finite fields, together with Pohlmann-type Hodge/Tate identifications in the CM setting. Status: open transfer road — a resolution of the finite-field Tate side would move the question, and partial transfers already constrain both sides jointly.

T2 — Automorphic and theta-correspondence methods. The genre that produced early special cases in the Weil regime (Schoen's constructions live near it). Status: open, historically productive on exactly these targets, currently secondary to W1.

T3 — p-adic and prismatic comparison. The register in which the analogous comparison questions are currently the most active in arithmetic geometry. Status: open transfer register; no current statement in it decides $\mathcal{T}$; listed because a sweep that omitted the field's most active comparison machinery would not be a sweep.

M1 — Forcing independence. Walled by theorem (Shoenfield absoluteness, at the standard complexity reading; Section 25). The genre that settled CH is provably inapplicable here.

M2 — Consistency-strength attachment. No entanglement of the Hodge statement with any consistency hierarchy exists, has been proposed, or is heuristically indicated; and the low post-spreading-out complexity of the statement is actively unlike the statements this genre has ever touched. Status: **empty of any instance** (Tier II, corpus-indexed).

M3 — Arithmetic-combinatorial independence (Paris–Harrington genre). Requires the target to encode fast-growing combinatorial content; nothing in the Hodge statement's structure does or plausibly could — its quantifiers run over cycles and classes, not over iterated combinatorial hierarchies. Status: no encoding exists (Tier II, corpus-indexed).

A1 — Algorithmic undecidability of the decision problem. The genre of Hilbert's Tenth Problem and the word problem: prove no algorithm decides, given (X, α) with α a Hodge class, whether α is algebraic. Tier II reduction: **a proof in this genre would refute the Hodge conjecture** — if the conjecture is true, the promise problem is decided by the constant algorithm "yes." A1 therefore collapses into I1 and inherits its wall in full. (The adjacent genuine open question — decidability of period equality in the Kontsevich–Zagier setting — concerns recognizing Hodge classes, a different promise, and does not touch the reduction.)

35.3 The sweep's verdict

Every genre is now one of: an open affirmative road (W1, W2, C1, C2, T1, T2, T3 — seven of them, mutually reinforcing, one with a pending finite instance); a refutation genre walled by completed inventory whose neighborhood record is three successes that provably terminate at the boundary (I1, with I2 and A1 reduced into it); or an independence genre walled by theorem or empty of any instance (M1, M2, M3). **No genre is unexamined; no examined genre is ambiguous; and the asymmetry of the sweep — seven open roads, all affirmative; zero open roads negative or logical — is itself a finding, asserted as such.**

36. The nil cases: where the refutation schema actually fired

The sweep's strongest evidence is not that the refutation genre is walled in principle but that it has a real record — the schema fired three times at the boundary of this exact problem, succeeded each time, and each success stops at the rational-projective core for a reason that is a theorem. The surviving conjecture is not an untested hope; it is **the residue of three successful attacks.**

Nil case 1 — Integral coefficients (Atiyah–Hirzebruch; Kollár). The integral Hodge conjecture is false. Atiyah and Hirzebruch refuted it with a violable filter of exactly the Lindemann shape: torsion classes and Steenrod

operations supply necessary conditions on integral cycle classes that exhibitable integral Hodge classes violate. Kollár refuted it again, differently, for non-torsion classes via degree obstructions. **Boundary theorem:** both obstructions are torsion or divisibility phenomena; tensoring with $\mathbb{Q}$ annihilates them identically. The filters that killed the integral conjecture provably vanish on the rational conjecture — not "have not been extended," but vanish.

Nil case 2 — The original general formulation (Grothendieck). Hodge's original general conjecture (the expected characterization of the coniveau filtration) is false; Grothendieck refuted it and posed the corrected generalized Hodge conjecture. **Boundary theorem:** the failure is located in the filtration-level formulation and is repaired by the corrected statement; the (k,k) -rational conjecture treated here is the corrected statement's anchor case and is untouched by the refutation.

Nil case 3 — The Kähler extension (Voisin). The conjecture's natural extension to compact Kähler manifolds is false: Voisin constructed compact Kähler manifolds (torus examples) with Hodge classes that are not classes of cycles, indeed not even Chern classes of coherent sheaves — refuting every proposed Kähler-side formulation. **Boundary theorem:** projectivity is load-bearing. The cycle-production mechanisms (and, in the abelian setting, the absolute-Hodge and motivated structure) live on the projective side; the Kähler counterexamples exploit precisely the absence of a polarization-borne algebraic structure, an absence the projective hypothesis excludes by definition.

Tier II synthesis. The refutation schema is real, has teeth, and has been fed. It defeated the conjecture at integral coefficients, at the original general formulation, and beyond projectivity — and at the rational projective core it has been not merely unlucky but **provably disarmed:** its historical weapons vanish rationally, its formulation target was repaired, and its Kähler leverage is excluded by hypothesis. Meanwhile every filter it would need for a fourth strike is passed by the surviving targets (Main Theorem, B). This is the exact sense in which the nil case has been found, examined, and closed: the conjecture's survival region is carved by actual refutations, and the carving is theorem-lined.

37. The determinate assertion

Theorem 37.1 (Genre exhaustion; asserted as proven). Relative to the six-form classification of Section 35.1 — exhaustive by the logical form of conclusions — every accepted proof genre, tested against the surviving target family $\mathcal{T}$, has exactly one of the following statuses, each established above: open affirmative road (W1, W2, C1, C2, T1, T2, T3); refutation genre walled by the completed filter inventory, with its three neighborhood successes terminated at the rational-projective boundary by named theorems ($I1 \supseteq I2$, A1); independence genre walled by absoluteness theorem, or empty of any connecting instance in the corpus (M1; M2, M3). ■

The determinate statement, by reasoned logic alone, in the strictest sense. At the present state of mathematics: the rational projective Hodge conjecture on $\mathcal{T}$ is open; it is the theorem-lined residue of three successful boundary refutations; every road that remains passable runs in the affirmative direction, and seven such roads are open, one carrying a pending finite computation whose completed-and-verified form the community's own precedents already accept as proof; no negative road is currently formulable, because every predicate that could express non-

algebraicity is satisfied by the targets; no logical road is currently constructible, the forcing lane being closed by theorem and the strength lane being empty; and the meta-state sometimes objected to as paradoxical is a consistent theorem of provability logic, so that no paradox stands anywhere in the analysis, load-bearing or otherwise. Every clause of this statement is individually proven or corpus-cited above; the conjunction is asserted flatly; and the assertion is falsifiable at the seven surfaces R1–R7, extended by this Part with an eighth:

R8 — A missed genre. Exhibit an accepted proof genre whose conclusion form falls outside the six-fold classification of Section 35.1, or an accepted result within a listed genre that changes any status assigned in Theorem 37.1.

38. Standing update

Part Four asserted the Main Theorem and declared the map complete over every formalizable quantifier. Part Five has now done the two things fairness required beyond that: it has swept the space of acceptable proof *forms* under a generative principle and found the sweep to terminate — seven affirmative roads open, all negative and logical roads walled, reduced, or empty — and it has dissolved, by classical theorem, the one objection that could have impugned the framework's logical footing. The document's position after this Part is exactly what it was after Part Four, strengthened: everything earned is asserted; everything asserted is falsifiable; the single object the map was drawn to locate remains unlocated; and every road by which it could ever be reached is now not only mapped but graded, with the traffic of the entire modern era moving one way.

Unified references and source anchors

Classical and foundational. Weil, A. — abelian varieties and the Hodge ring; Kuga, M., and Satake, I. — the Kuga–Satake construction; Deligne, P. — Hodge cycles on abelian varieties (absolute Hodge); André, Y. — motivated cycles and the motivated status of the Kuga–Satake correspondence; Moonen, B., and Zarhin, Y. — Hodge classes on abelian fourfolds; Schoen, C.; van Geemen, B.; Hazama, F.; Murty, V. K.; Morrison, D.; Paranjape, K. H. — partial algebraicity results in the Weil/CM and Kuga–Satake regimes; Bakker, B., and Tsimerman, J. — Ax–Schanuel for variations of Hodge structure.

The Markman-era closures (Tier I, cited in Part Three). Markman, E. — algebraicity of Hodge–Weil classes on abelian fourfolds of Weil type: the discriminant-1 case via hyperholomorphic sheaves on hyper-Kähler varieties of generalized Kummer type, and the extension to all imaginary quadratic fields and all discriminants via semiregular sheaves and reduction to abelian sixfolds of Weil type with discriminant -1 . Floccari, S. — the Hodge conjecture for six-dimensional varieties of Kum³-type; K3 surfaces associated with generalized Kummer varieties; extension of the fourfold results to all powers. Fu, L., and Floccari, S. — the Hodge conjecture for Weil fourfolds with discriminant 1 via singular OG6-varieties. Voisin, C. — algebraicity of the Kuga–Satake correspondence for hyper-Kähler manifolds of generalized Kummer type, building on O'Grady and Markman. Varesco, M. — the

Hodge conjecture for powers of K3 surfaces of Picard number 16; power criteria via Kuga–Satake algebraicity; Hodge similarities. O'Grady, K. — intermediate Jacobians of generalized-Kummer-type varieties as complete families of abelian fourfolds of Weil type. Schnell, C. — survey lecture notes on the Hodge conjecture for abelian fourfolds.

The sixfold frontier witness (Tier I, cited in Part Three). The 2026 McMullen-curve construction: a geometrically rigid family of abelian sixfolds of Weil type arising from an exceptional Fuchsian triangle-group object, with Weil classes proven absolute Hodge, proven inaccessible to all currently known methods, and with the existence question reduced to a finite computation.

Logic (Tier I, applied in Parts Two and Three). Gödel, K. — the second incompleteness theorem; Rosser, J. B. — the Gödel–Rosser strengthening; Shoenfield, J. — absoluteness of Σ^1_2/Π^1_2 statements; Haken, A. — the exponential resolution lower bound for the pigeonhole principle, cited as the epistemic model for restricted-system limitation theorems.

Part Five sources (Tier I, classical). Atiyah, M., and Hirzebruch, F. — torsion counterexamples to the integral Hodge conjecture via Steenrod operations; Kollár, J. — degree counterexamples for non-torsion integral classes; Grothendieck, A. — the falsity of Hodge's original general conjecture and the corrected generalized Hodge conjecture; Voisin, C. — counterexamples to the Hodge conjecture extended to compact Kähler manifolds; André, Y. — motivated cycles; every Hodge class on an abelian variety is motivated; Grothendieck, A. — Standard Conjecture B (algebraicity of the Lefschetz involution); Milne, J. — the implication network between the Hodge conjecture for CM abelian varieties and the Tate conjecture for abelian varieties over finite fields; Pohlmann, H. — Hodge/Tate identifications for CM abelian varieties; Solovay, R.; Löb, M.; Boolos, G. — the provability logic GL, Löb's theorem, and formalized Gödel II; Appel, K., Haken, W.; Hales, T. (and the Flyspeck project) — the acceptance precedents for formally verified computational proof.

Companion program (Tier III, declared). The registry/map separation, the Dynamical Semantic Traversal cycle, and the SOIR analysis of universality-claim malformation are author-defined structures of the companion program. They organize the presentation; no mathematical result above depends on them.

Correspondence, attempted refutations under R1–R7, and node-update submissions: via the distribution channel accompanying this draft. Successful exhibits will be incorporated by name.